Constrained Stochastic Policy Targeting under Heterogeneous Partial Interference

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Abstract

Under heterogeneous interference, treating individuals with particular characteristics affects the population average outcome through their heterogeneous direct response to treatment and indirect effect on others, creating opportunities for efficient policy design. For instance, policymakers may minimize resource use by vaccinating individuals identified as superspreaders to achieve a target reduction in disease incidence. In such setting, we first study the problem of learning individualized stochastic treatment policies that minimize population average treatment probability subject to a prespecified outcome target. We propose an estimation approach based on a smooth surrogate for constraint violations and interference-aware inverse-probability-weighting or self-normalized estimators of the population average outcome, and establish finite-sample regret guarantees for both estimators. We also consider the complementary problem of maximizing the population average outcome subject to a constraint on average treatment probability. We apply the proposed methodology to a Colombian conditional cash transfer program aimed at improving school attendance. The application demonstrates that accounting for interference and individualizing treatment probabilities can reduce resource use relative to policies that ignore interference or impose homogeneous treatment probabilities.

Keywords: Causal inference, optimal treatment policy, spillovers.

1 Introduction

Many interventions are implemented in settings in which individuals interact, so that the treatment assigned to one individual may affect not only that individual's outcome but also the outcomes of others. Such phenomenon, known as interference, arises naturally within households, schools, villages, and other socially or physically connected groups (Cox, 1958). Accounting for these dependencies is particularly important for optimal policy learning, aimed at learning treatment rules that optimize a population-level objective (Athey and Wager, 2021; Kitagawa and Tetenov, 2018; Zhao et al., 2012; Qian and Murphy, 2011). Without interference, individualized treatment rules exploit heterogeneity in individuals' responses to their own treatment. Under interference, heterogeneity can also arise in the effects that one individual's treatment has on others, which may depend on their characteristics and those of connected individuals and on the interaction structure in the population (Bargagli-Stoffi et al., 2025; Dean et al., 2025; Viviano, 2024). For instance, in the Colombian conditional cash transfer program *Subsidios Condicionados a la Asistencia Escolar*, prior work documents both heterogeneity in treatment response by gender and socioeconomic status and heterogeneous adverse spillovers on untreated siblings, with larger attendance reductions among female siblings of treated students (Barrera-Osorio et al., 2011; Ferreira et al., 2009).

In such settings, accounting for interference is consequential for policy design for two reasons. First, a policy-learning procedure that disregards interference can lead to biased estimates of relevant policy quantities and select an allocation that is suboptimal for the population objective (Viviano, 2024; Zhang and Imai, 2024). Second, heterogeneity in direct and spillover effects can be leveraged to design efficient policy targeting when resources are scarce. If spillovers are beneficial, prioritizing individuals whose treatment generates larger gains for others can attain a prespecified outcome using fewer resources or increase population welfare for a given treatment budget. If spillovers are harmful, an interference-aware policy can instead avoid allocations that generate negative indirect effects on others' outcomes.

Most of the policy-learning literature considers the problem of assigning treatment to maximize population welfare, possibly subject to a constraint on available resources (Kosorok and Moodie, 2015; Kitagawa and Tetenov, 2018; Athey and Wager, 2021). A complementary problem arises when the policymaker instead faces a prespecified outcome target and seeks the least resource-intensive policy that attains it (Bhattacharya and Dupas, 2012). In this paper, we consider both problems under partial interference, where interference may occur within given clusters but not across clusters. Specifically, let $p \in \mathcal{P}$ denote a covariate-dependent treatment policy, $\overline{Y}(p)$ the population average outcome induced by the policy, and $\overline{C}(p)$ its population average treatment intensity. We study $\max_{p\in\mathcal{P}} \overline{Y}(p)$ subject to

$\overline{C}(p) \leq B$, and $\min_{p \in \mathcal{P}} \overline{C}(p)$ subject to $\overline{Y}(p) \geq T$. The first formulation, which we refer to as the "resource-constrained" problem, is a version of the familiar welfare-maximization problem under a resource budget. In contrast, the second formulation, referred to as the "outcome-target" problem, selects the least intensive treatment policy that attains a pre-specified level of population outcome. Although understudied, the outcome-target problem is particularly relevant when policymakers face externally specified performance goals, such as the UN Sustainable Development Goals and WHO targets for malaria reduction (United Nations, 2016; World Health Organization, 2021). For instance, in the Colombian cash transfer program, a policymaker may seek the least intensive cash-transfer policy that attains a specified level of school attendance. The logic is analogous to sample-size planning, where a desired level of statistical power is specified first and the required sample size is then determined. In this paper, our primary focus is the outcome-target problem; we then extend the analysis to the resource-constrained problem.

Under partial interference, policy evaluation involves high-dimensional cluster-level treatment assignments. This poses challenges for deterministic policies, since inverse-probability-weighted (IPW) evaluation requires exact matches between observed and policy-prescribed assignments, which can be rare in finite samples (Zhang and Imai, 2024). We therefore consider individualized stochastic treatment policies that assign treatment probabilities based on individual- and cluster-level covariates. Such policies allow multiple observed treatment configurations to contribute to estimation (Kennedy, 2019; Swaminathan and Joachims, 2015a) and, for smooth parametric classes, allow for efficient gradient-based optimization (Luedtke and van der Laan, 2016; Fang et al., 2023). Moreover, under interference, stochastic policies can be optimal and strictly outperform deterministic allocations (Dal Torrione et al., 2026)[1].

The estimated constraint creates a challenge absent from standard unconstrained policy learning, since sampling error can make an infeasible policy appear feasible (Sun, 2026). To address this problem, we develop an empirical policy estimator that tightens the estimated constraint through a margin ϵ and uses a smooth surrogate for constraint violation. We establish non-asymptotic regret guarantees for the resulting IPW estimator, where the regret bound balances the surrogate approximation error with the objective and constraint estimation error, yielding a regret of order $N^{-1/4}$. Under additional conditions, we also obtain the

[1] Bickel and Doksum (2015, Chapter 1.3) establish that randomized decision rules strictly outperform deterministic ones when the risk set of the deterministic rules is non-convex. Similarly, under interference, stochastic policies outperform deterministic ones when the outcome space is non-convex due to interactions. For instance, consider a cluster of units with identical covariates where a mixed treatment allocation yields a greater cluster-average outcome than uniform allocations. Because any deterministic covariate-based rule is constrained to treat either all units or no units within the cluster, deterministic rules yield smaller average outcomes, whereas an independent stochastic policy achieves a greater expected outcome by convexifying the action space.

sharper rate $O\{\sqrt{\log(N)/N}\}$.

Under interference, IPW estimators may suffer from extreme weights leading to high variance due to the reweighting of joint treatment configurations at the cluster level. Furthermore, because this variance profile can differ substantially across policies, enforcing a uniform margin can be excessively conservative. We address both issues in two complementary ways. First, alongside the interference-aware IPW estimator, we consider its self-normalized, or Hájek, analogue (Hájek, 1971; Tchetgen Tchetgen and VanderWeele, 2012; Papadogeorgou et al., 2019), which can stabilize cluster-level weights. Second, drawing on empirical Bernstein arguments (Maurer and Pontil, 2009; Swaminathan and Joachims, 2015a), we develop policy-specific, variance-adaptive constraint margins for both IPW and Hájek estimation, which impose stricter feasibility requirements on policies with more variable outcome estimates and less conservative requirements on those estimated more precisely. We establish finite-sample regret guarantees for the Hájek estimator, which attain the same rates as the IPW estimator, and for the variance-adaptive procedures under both IPW and Hájek estimation. To our knowledge, these are the first uniform empirical Bernstein bounds for interference-aware IPW and Hájek policy evaluation, and may be of independent statistical interest.

Finally, we study the finite-sample performance of the proposed methods in simulations and apply them to a Colombian conditional cash transfer experiment with potential within-household interference.

Related Literature Our work bridges the two distinct strands of literature concerning causal inference under interference and optimal treatment policy learning. An extensive body of work on interference develops identification and estimation methods for direct, spillover, and overall causal effects (Hudgens and Halloran, 2008; Tchetgen Tchetgen and VanderWeele, 2012; Aronow and Samii, 2017; Forastiere et al., 2021; Leung, 2020; Dal Torrione et al., 2025; Park and Kang, 2022), while also exploring effect heterogeneity and covariate-dependent allocation strategies (Bargagli-Stoffi et al., 2025; Dean et al., 2025; Papadogeorgou et al., 2019). The policy-learning literature (Athey and Wager, 2021; Kitagawa and Tetenov, 2018; Zhao et al., 2012; Qian and Murphy, 2011) operates predominantly under the assumption of no interference. While an emerging domain intersects these fields by studying treatment allocation and policy learning under interference (Kitagawa and Wang, 2023; Ananth, 2024; Viviano, 2024; Viviano and Rudder, 2024; Zhang and Imai, 2024), most existing approaches focus on welfare maximization and deterministic treatment rules, often imposing restrictive structural assumptions on the interference mechanism to maintain tractability. We instead study constrained policy learning with stochastic policies under partial interference, allowing

potential outcomes to depend arbitrarily on the treatment assignments of all units within a cluster.

Our outcome-target formulation is most closely related to Park et al. (2024), who introduce minimum resource threshold policies determining the minimum proportion of treated units required for a cluster to attain a cluster-level outcome target. We instead aim to learn individualized treatment probabilities as functions of observed characteristics, while imposing the constraint on the population average outcome. In no-interference settings, Bhattacharya and Dupas (2012) characterize budget-constrained allocations across unrestricted policy classes using individual effect-to-cost ratios; by contrast, we learn the optimal policy a prespecified class while explicitly allowing for spillover effects. More recently, Sun (2026) studies welfare maximization under an estimated budget constraint for deterministic eligibility policies, deriving asymptotic results on welfare efficiency and feasibility. We instead consider stochastic policies under interference, minimizing treatment intensity subject to an outcome target, and derive finite-sample regret guarantees for interference-aware estimators. Finally, our work relates to self-normalized off-policy estimation, which has received considerable attention in the machine-learning literature (e.g., Swaminathan and Joachims, 2015b; Kuzborskij et al., 2021). We contribute uniform finite-sample empirical Bernstein bounds for interference-aware IPW and Hájek estimators over potentially infinite policy classes, together with their implications for constrained policy regret.

Organization of the paper. Section 2 introduces the partial-interference framework, stochastic policy class, and outcome-target optimization problem. Section 3 develops our proposed policy estimators. Section 4 establishes the regret guarantees, and Section 5 develops the variance-adaptive constraint margins. Section 6 discusses the complementary resource-constrained problem of maximizing welfare subject to a constraint on the average treatment probability. Sections 7 and 8 present the simulation study and the empirical application, and Section 9 concludes. Appendix A provides the detailed analysis of the resource-constrained problem, Appendix B reports additional tables for the empirical application, and Appendix C contains the proofs.

2 Framework

2.1 Setup and Notation

Let N be the total number of clusters, with each cluster indexed by $i = 1, \ldots, N$. For each cluster i, n_i is the number of units and units are indexed by $j = 1, \ldots, n_i$, such that the subscript ij denotes unit j in cluster i. For each unit j in cluster i we observe

the treatment assignment $A_{ij} \in \{0,1\}$, the outcome $Y_{ij} \in \mathbb{R}$ and K individual covariates $\boldsymbol{X}_{ij} \in \mathcal{X}$, with $\mathcal{X} \subseteq \mathbb{R}^K$. In addition, each cluster carries L cluster-level covariates $\boldsymbol{V}_i \in \mathcal{V} \subseteq \mathbb{R}^L$. Without loss of generality, we assume that a larger value of Y_{ij} is preferred. Finally, for each cluster i, we use $\boldsymbol{A}_i \in \mathcal{A}(n_i) = \{0,1\}^{n_i}$, $\boldsymbol{X}_i = (\boldsymbol{X}_{i1}\ \boldsymbol{X}_{i2}\ \cdots\ \boldsymbol{X}_{in_i})^\top$, and $\boldsymbol{Y}_i = (Y_{i1}\ Y_{i2}\ \cdots\ Y_{in_i})$ to denote the vector of treatment, the $n_i \times K$ matrix of covariates, and the vector of observed outcomes, respectively. Let us denote by $\boldsymbol{O}_i = \{\boldsymbol{Y}_i, \boldsymbol{A}_i, \boldsymbol{X}_i, \boldsymbol{V}_i\}$ the observed data in the cluster. We assume that $\{\boldsymbol{O}_i\}_{i=1}^N$ is an *i.i.d.* sample drawn from a superpopulation of clusters characterized by (unknown) distribution G_0. Throughout, lower case letters indicate realizations of the corresponding random variables.

In the presence of interference, the potential outcome of an individual may depend on the treatment of other individuals. In full generality, let $Y_{ij}(\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_N)$ denote the potential outcome for unit j in cluster i when the treatment of units across all clusters is equal to $(\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_N)$ for $\boldsymbol{a}_i \in \mathcal{A}(n_i)$. Throughout the paper, we make a partial (clustered) interference assumption, which restricts interference to occur only within clusters but not across clusters (Sobel, 2006; Hudgens and Halloran, 2008). Formally,

Assumption 1 (Partial Interference)**.** *For all $i = 1, \ldots, N$ and $j = 1, \ldots, n_i$, and for all vector of cluster treatment assignments $(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_N)$ and $(\boldsymbol{a}'_1, \ldots, \boldsymbol{a}'_N)$ such that $\boldsymbol{a}_i = \boldsymbol{a}'_i$, we have that $Y_{ij}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_N) = Y_{ij}(\boldsymbol{a}'_1, \ldots, \boldsymbol{a}'_N)$.*

Under this assumption, for each unit j in cluster i, let $Y_{ij}(\boldsymbol{a}_i)$ denote the potential outcome under cluster treatment $\boldsymbol{a}_i$, representing the outcome that would be observed for unit j had the treatment vector in cluster i been set to $\boldsymbol{a}_i$.

2.2 Individualized Stochastic Treatment Policy

We consider stochastic, covariate-dependent treatment allocations where each individual is treated independently with a probability determined by their individual and cluster-level covariates. Specifically, we define an individualized stochastic treatment policy as a function from the individual and cluster-level space to the interval $[0,1]$, $p : \mathcal{X} \times \mathcal{V} \mapsto [0,1]$. Hence, $p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i) = \mathbb{P}_p(A_{ij} = 1 \mid \boldsymbol{X}_{ij}, \boldsymbol{V}_i)$ specifies the conditional probability of treatment for an individual with characteristics $(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)$. Since the policy imposes independent treatment assignment conditional on covariates, the probability of a treatment vector $\boldsymbol{a}_i \in \mathcal{A}(n_i)$ in cluster i under policy p is

$$\pi_i(\boldsymbol{a}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p) \ := \ \mathbb{P}_p(\boldsymbol{A}_i = \boldsymbol{a}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i) = \prod_{j=1}^{n_i} p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)^{a_{ij}} (1 - p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i))^{1-a_{ij}} . \quad (1)$$

We aim to choose an optimal policy p within a prespecified class of functions $\mathcal{P}$. We assume that $\mathcal{P}$ contains at least one constant policy. Possible function classes in the present

work include the class of linear probability functions $\mathcal{P}_{\Gamma,\text{lin}} = \{(\boldsymbol{x},\boldsymbol{v}) \mapsto \langle \gamma, (\boldsymbol{x},\boldsymbol{v})\rangle : \gamma \in \Gamma\}$ for some parameter space Γ, and the class of logistic probability functions $\mathcal{P}_{\Gamma,\text{logit}} = \{(\boldsymbol{x},\boldsymbol{v}) \mapsto \sigma(\langle \gamma, (\boldsymbol{x},\boldsymbol{v})\rangle) : \gamma \in \Gamma\}$ with $\sigma(u) = (1+\exp(-u))^{-1}$ (e.g. Swaminathan and Joachims, 2015a; Papadogeorgou et al., 2019; Dean et al., 2025).

2.3 Optimization Problem

Adapting the definition in Hudgens and Halloran (2008) and Tchetgen Tchetgen and VanderWeele (2012), we define the population average potential outcome under policy p as

$$\bar{Y}(p) = \mathbb{E}_{G_0}\left\{ \frac{1}{n_i} \sum_{j=1}^{n_i} \sum_{\boldsymbol{a}_i \in \mathcal{A}(n_i)} Y_{ij}(\boldsymbol{a}_i)\, \pi_i(\boldsymbol{a}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p) \right\}$$

Similarly we define the population average treatment probability as

$$\bar{C}(p) = \mathbb{E}_{G_0}\left\{ \frac{1}{n_i} \sum_{j=1}^{n_i} p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i) \right\}$$

which represents a measure of resource use of policy p. Our problem consists in finding a policy $p^\star$ in a class of policies $\mathcal{P}$ which minimizes the population treatment propensity while achieving a prespecified outcome level:

$$p^\star \in \arg\min_{p\in\mathcal{P}} \bar{C}(p) \qquad \text{s.t.} \quad \bar{Y}(p) \geq T. \tag{2}$$

Problem (2) can be conveniently rewritten in an equivalent unconstrained form as

$$p^\star \in \arg\min_{p\in\mathcal{P}} R_{G_0}(p), \qquad \text{where} \qquad R_{G_0}(p) := \bar{C}(p) + C_0 \cdot \mathbf{1}\{\bar{Y}(p) < T\}. \tag{3}$$

where C_0 is a constant larger than 1. We refer to $R_{G_0}(p)$ as the "risk" of policy p. This risk definition incorporates both our preference for minimizing resource use and the penalty associated with violating the outcome constraint.

The population average outcome is not identified from the observed data without further assumptions. Let $f(\boldsymbol{A}_i = \boldsymbol{a}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i)$ denote the known cluster treatment propensity score under G_0. For causal identification, we impose the following assumptions, which extend consistency, conditional unconfoundedness, and overlap to partial interference settings (e.g. Tchetgen Tchetgen and VanderWeele, 2012; Park and Kang, 2022).

Assumption 2. *For all $i = 1,\ldots,N$ and $j = 1,\ldots,n_i$ the following holds:*

a) Consistency: $\boldsymbol{Y}_i = \boldsymbol{Y}_i(\boldsymbol{A}_i)$.

b) Unconfoundedness: $\boldsymbol{Y}_i(\boldsymbol{a}_i) \perp\!\!\!\perp \boldsymbol{A}_i \,|\, \boldsymbol{X}_i, \boldsymbol{V}_i$ *for all* $\boldsymbol{a}_i \in \mathcal{A}(n_i)$.

c) Positivity: for some $\eta > 0$, $f(\boldsymbol{A}_i = \boldsymbol{a}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i) > \eta$ *for all* $\boldsymbol{a}_i \in \mathcal{A}(n_i)$ *such that* $\pi_i(\boldsymbol{a}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p) > 0$ *for some* $p \in \mathcal{P}$.

Under Assumptions 1 and 2, Problem (3) can be rewritten in terms of observed data as

$$p^\star \in \arg\min_{p\in\mathcal{P}} \left[\bar{C}(p) + C_0 \cdot \mathbf{1}\left\{\mathbb{E}_{G_0}\left\{\frac{1}{n_i}\sum_{j=1}^{n_i} Y_{ij} w_i(p)\right\} < T\right\}\right], \quad w_i(p) = \frac{\pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p)}{f(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i)} \tag{4}$$

where the weights $w_i(p)$ are the ratio of the probability of the observed cluster assignment under p to its probability under the data-generating assignment mechanism.

Since the risk depends on the unknown population distribution G_0, our goal is to estimate a policy $\hat{p}$ from the data. For this purpose, we use an empirical analog of the risk in the spirit of the Empirical Welfare Maximization framework (EWM), which maximizes a sample analog of the population welfare over a policy class (Kitagawa and Tetenov, 2018). As in that literature, a learned policy can be evaluated by the population risk it attains relative to the best attainable risk in $\mathcal{P}$, that is, by its regret. Accordingly, we define

$$R^\star_{G_0} := \inf_{p\in\mathcal{P}} R_{G_0}(p),$$

and we evaluate an estimated policy $\widehat{p}$ through its expected regret, defined as

$$\mathrm{Reg}(\hat{p}) = \mathbb{E}_{\mathcal{O}}[R_{G_0}(\widehat{p})] - R^\star_{G_0}. \tag{5}$$

where the expectation is taken over the distribution of the observed data $(\boldsymbol{O}_1, \ldots, \boldsymbol{O}_N)$, sampled i.i.d. from the population distribution G_0. Hence, given this performance criterion, in the following analysis we do not require $p^\star$ to be unique or $\hat{p}$ to converge to a specific minimizer of R_{G_0} over $\mathcal{P}$. For $t \in \mathbb{R}$, we further denote by

$$\bar{C}^\star(t) := \inf_{p\in\mathcal{P}} \left\{\bar{C}(p) : \ \bar{Y}(p) \geq t\right\}$$

the minimal resource use required to attain outcome level t within $\mathcal{P}$. Since $\bar{C}(p) \leq 1 < C_0$, whenever some $p \in \mathcal{P}$ satisfies $\bar{Y}(p) \geq T$ we have $R^\star_{G_0} = \bar{C}^\star(T)$.

3 Estimation and Optimization

To estimate the optimal policy, one could replace the population quantities in (3) with their empirical analogues and minimize the resulting criterion. However, the indicator for constraint violation makes the objective discontinuous in the policy, complicating both theoretical analysis and numerical optimization. To this end, we replace the indicator with a

smooth surrogate. For $\epsilon > 0$, define

$$h_\epsilon(u;T) = \begin{cases} 1, & u \le T, \\ 1 - \dfrac{2}{\epsilon^2}(u-T)^2, & T < u < T + \dfrac{\epsilon}{2}, \\ \dfrac{2}{\epsilon^2}(T+\epsilon-u)^2, & T + \dfrac{\epsilon}{2} \le u < T+\epsilon, \\ 0, & u \ge T+\epsilon. \end{cases} \tag{6}$$

The function $h_\epsilon(u;T)$ upper bounds the constraint-violation indicator $\mathbf{1}(u < T)$, equals one for $u \le T$, and vanishes for $u \ge T + \epsilon$. The parameter ϵ determines the width of the buffer region $[T, T+\epsilon]$, over which the penalty decreases smoothly from one to zero. Moreover, $h_\epsilon(u;T) \to \mathbf{1}(u < T)$ pointwise for every $u \ne T$ as $\epsilon \to 0$. Unlike the truncated linear hinge, $h_\epsilon(\cdot;T)$ is continuously differentiable and $2/\epsilon$-Lipschitz. Using this surrogate, we consider

$$\min_{p\in\mathcal{P}} V_{\epsilon,G_0}(p), \qquad V_{\epsilon,G_0}(p) \;=\; \bar{C}(p) + C_0\, h_\epsilon\big(\bar{Y}(p);T\big), \tag{7}$$

and refer to $V_{\epsilon,G_0}(p)$ as the surrogate risk. We denote by $V^\star_{\epsilon,G_0} := \inf_{p\in\mathcal{P}} V_{\epsilon,G_0}(p)$ the optimal surrogate risk and by $p^\star_\epsilon \in \arg\min_{p\in\mathcal{P}} V_{\epsilon,G_0}(p)$ a surrogate-optimal policy. The surrogate risk assigns penalty C_0 to infeasible policies with $\bar{Y}(p) < T$, zero penalty to policies satisfying the constraint with a margin, that is $\bar{Y}(p) \ge T + \epsilon$, and an intermediate penalty to policies with $\bar{Y}(p) \in [T, T+\epsilon]$. Since $h_\epsilon(u;T) \ge \mathbf{1}(u < T)$, the surrogate constraint provides a conservative approximation to the original constraint, so $V_{\epsilon,G_0}(p) \ge R_{G_0}(p)$ for every $p \in \mathcal{P}$; in particular, at a surrogate-optimal solution, $V_{\epsilon,G_0}(p^\star_\epsilon) \ge R_{G_0}(p^\star_\epsilon)$. Whenever there exists a policy $p^0_\epsilon \in \mathcal{P}$ such that $\bar{Y}(p^0_\epsilon) \ge T + \epsilon$, we have $V_{\epsilon,G_0}(p^0_\epsilon) = \bar{C}(p^0_\epsilon) \le 1 < C_0$, whereas every infeasible policy has surrogate risk at least C_0. Hence an optimal $p^\star_\epsilon$ is feasible for the original constraint, $\bar{Y}(p^\star_\epsilon) \ge T$, and under this condition the following also holds:

$$V^\star_{\epsilon,G_0} = V_{\epsilon,G_0}(p^\star_\epsilon) \ge R_{G_0}(p^\star_\epsilon) = \bar{C}(p^\star_\epsilon) \ge R^\star_{G_0} = \bar{C}^\star(T),$$

so that $V^\star_{\epsilon,G_0} \ge R^\star_{G_0}$ and $\bar{C}(p^\star_\epsilon) \ge \bar{C}^\star(T)$: the surrogate-optimal policy uses, on average, at least as much resource as an optimal policy for the original problem.

3.1 Empirical Objective

The average treatment probability $\bar{C}(p)$ is estimated by its sample average, given by

$$\widehat{\bar{C}}(p) = \frac{1}{N}\sum_{i=1}^{N} \bar{C}_i(p), \qquad \bar{C}_i(p) = \frac{1}{n_i}\sum_{j=1}^{n_i} p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i).$$

For the outcome constraint, we consider two estimators. We first define the interference-aware IPW estimator, $\widehat{\bar{Y}}_{\text{I-IPW}}(p)$, as follows:

$$\widehat{\bar{Y}}_{\text{I-IPW}}(p) \;=\; \frac{1}{N}\sum_{i=1}^{N}\psi_i^{\text{I-IPW}}(p), \qquad \text{with } \psi_i^{\text{I-IPW}}(p) \;=\; \frac{1}{n_i}\sum_{j=1}^{n_i} Y_{ij}\frac{\pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p)}{f(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i)}, \tag{8}$$

The IPW estimator weights the average observed outcomes within clusters by the ratio of the probability of the observed cluster-level observed treatment assignment under the target policy to that under the observed treatment mechanism, given by the $f(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i)$, which, throughout, we assume to be known. By (4), $\mathbb{E}_{G_0}\{\psi_i^{\text{I-IPW}}(p)\} = \bar{Y}(p)$, so $\widehat{\bar{Y}}_{\text{I-IPW}}(p)$ is unbiased for the population average outcome under policy p under Assumptions 1 and 2 (Tchetgen Tchetgen and VanderWeele, 2012; Papadogeorgou et al., 2019).

The IPW estimator in (8) can be highly variable especially when n_i is large. As a potentially more stable alternative, we therefore also consider the self-normalized, or Hájek, estimator, $\widehat{\bar{Y}}_{\text{I-HJ}}(p)$, defined as follows

$$\widehat{\bar{Y}}_{\text{I-HJ}}(p) = \frac{N^{-1}\sum_{i=1}^{N}\bar{Y}_i\, w_i(p)}{D_N(p)}, \qquad \text{with } D_N(p) = \frac{1}{N}\sum_{i=1}^{N} w_i(p), \tag{9}$$

where $\bar{Y}_i = n_i^{-1}\sum_{j=1}^{n_i} Y_{ij}$ is the observed cluster-average outcome. Similar cluster-level weights have been used previously in the interference literature (Papadogeorgou et al., 2019; Dean et al., 2025). Although the Hájek estimator is generally biased in finite samples, it is consistent for the population average outcome, and the weight standardization can yield efficiency gains relative to the unstandardized IPW estimator (Hájek, 1971; Liu et al., 2016). We propose the following optimal policy estimators:

$$\widehat{p}_{\text{I-IPW}} \in \arg\min_{p\in\mathcal{P}} \widehat{V}_{\epsilon,\text{I-IPW}}(p), \qquad \widehat{p}_{\text{I-HJ}} \in \arg\min_{p\in\mathcal{P}} \widehat{V}_{\epsilon,\text{I-HJ}}(p).$$

with

$$\widehat{V}_{\epsilon,\text{I-IPW}}(p) = \widehat{\bar{C}}(p) + C_0\, h_\epsilon\big(\widehat{\bar{Y}}_{\text{I-IPW}}(p); T\big), \qquad \widehat{V}_{\epsilon,\text{I-HJ}}(p) = \widehat{\bar{C}}(p) + C_0\, h_\epsilon\big(\widehat{\bar{Y}}_{\text{I-HJ}}(p); T\big), \tag{10}$$

Both estimators $\widehat{p}_{\text{I-IPW}}$ and $\widehat{p}_{\text{I-HJ}}$ minimize the empirical analogue of $V_{\epsilon,G_0}(p)$, denoted by $\widehat{V}_{\epsilon,\text{I-IPW}}(p)$ and $\widehat{V}_{\epsilon,\text{I-HJ}}(p)$, obtained by replacing $\bar{C}(p)$ with the sample average treatment propensity $\widehat{\bar{C}}(p)$, and $\bar{Y}(p)$ with the corresponding population average outcome estimator, $\widehat{\bar{Y}}_{\text{I-IPW}}(p)$ and $\widehat{\bar{Y}}_{\text{I-HJ}}(p)$. The smooth surrogate, together with the use of stochastic policies, permits efficient gradient-based optimization. In Sections 7 and 8, we minimize (10) using the L-BFGS-B algorithm (Liu and Nocedal, 1989), as implemented in the R `optim` function.

4 Regret Analysis

To establish the theoretical properties of our policy estimators, we make the following assumptions.

Assumption 3. *The following holds for all $i = 1, \ldots, N$ and $j = 1, \ldots, n_i$:*

a) Bounded outcomes: there exists a constant M such that $|Y_{ij}(\boldsymbol{a}_i)| < M$ for all $\boldsymbol{a}_i$.

b) Bounded cluster size: there exists a constant n_{max} such that $n_i < n_{max}$.

Assumption 3.a) assumes bounded outcomes and is standard in the literature (Kitagawa and Tetenov, 2018; Zhang and Imai, 2024). Assumption 3.b) imposes bounded cluster size n_i, ensuring that the cluster size is small relative to the number of clusters N under the asymptotic regime where $N \to \infty$.

4.1 Regret for the IPW Estimator

Our goal is to bound the regret of $\widehat{p}_{\text{I-IPW}}$, $\text{Reg}(\widehat{p}_{\text{I-IPW}})$, as defined in Eq. (5).

Theorem 1. *Under Assumptions 1–3,*

$$\text{Reg}(\widehat{p}_{\text{I-IPW}}) \leq \underbrace{V^{\star}_{\epsilon,G_0} - R^{\star}_{G_0}}_{(a)} + \underbrace{8 n_{max} \mathcal{R}_N(\mathcal{P})}_{(b)} + \underbrace{8\sqrt{2}\, n_{max}^{3/2} \frac{M}{\eta} \frac{C_0}{\epsilon} \mathcal{R}_N(\mathcal{P}) + \frac{4 C_0 M}{\eta \epsilon} \sqrt{\frac{2 \log 2}{N}}}_{(c)} .$$

where $\mathcal{R}_N(\mathcal{P})$ denotes the expected Rademacher complexity of the function class $\mathcal{P}$, which is a measure of complexity of $\mathcal{P}$ [2].

The upper bound on the regret depends on three terms. Term (a) is the approximation error arising from using the surrogate (approximate) risk instead of the original problem, and it is given by the difference between the optimal surrogate risk and the optimal risk over $\mathcal{P}$. In other words, this term represents the optimality gap induced by the use of the surrogate loss function. Terms (b) and (c) control the estimation error of the surrogate risk, $\mathbb{E}_{\mathcal{O}}[V_{\epsilon,G_0}(\widehat{p}_{\text{I-IPW}}) - V^{\star}_{\epsilon,G_0}]$, via the uniform estimation errors $\sup_{p \in \mathcal{P}} |\widehat{\bar{C}}(p) - \bar{C}(p)|$ and $\sup_{p \in \mathcal{P}} |\widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p)|$, respectively. In particular, term (c) depends linearly on the expected Rademacher complexity of the policy class multiplied by $n_{\max}^{3/2}$, and is inversely

[2] Specifically, given iid data $Z_1, ..., Z_N$, $Z_i \in \mathcal{Z}$ and a set of functions $\mathcal{F}$ where $f : \mathcal{Z} \to \mathbb{R}$, the expected Rademacher complexity is defined as

$$\mathcal{R}_N(\mathcal{F}) = \mathbb{E}_{\mathcal{Z},\sigma}\Big[\sup_f \frac{1}{N} \sum_i^N \sigma_i f(\boldsymbol{Z}_i)\Big]$$

where $\sigma_1, ..., \sigma_N \in \{-1, 1\}$ are i.i.d. random variables with $\mathbb{P}[\sigma_i = 1] = \mathbb{P}[\sigma_i = -1] = 1/2$ (see def. (26.4) in Shalev-Shwartz and Ben-David, 2014).

proportional to the lower bound of the propensity score η and the margin parameter ϵ. The convergence properties of this upper bound depend on the convergence to zero of the estimation error and the approximation error. For linear probability policies with bounded covariates and coefficients, $\mathcal{R}_N(\mathcal{P}_{\Gamma,\text{lin}}) = O(N^{-1/2})$. The same rate holds for the class of logistic policies $\mathcal{P}_{\Gamma,\text{logit}}$ under the same conditions.[3] Thus, whenever $\mathcal{R}_N(\mathcal{P}) = O(N^{-1/2})$, the estimation error is $O(\epsilon^{-1}N^{-1/2})$. Convergence of the regret bound additionally requires the approximation error to vanish.

On the other hand, the behavior of the approximation error depends only on the margin parameter ϵ. Ideally, if $\epsilon \to 0$, we should expect the approximation error to approach zero as well. Below, we show that, if there exists a direction of growth in the outcome, the approximation error is, in fact, $O(\epsilon)$.

Theorem 2 (Approximation error)**.** *For some $p^\star \in \arg\min_{p\in\mathcal{P}} R_{G_0}(p)$, $\bar{C}(p)$ is Lipschitz around $p^\star$, with respect to the norm $\|\cdot\|$, with Lipschitz constant L_p. Additionally, there exists a direction d and positive constants κ and t_0 such that $p^\star + td \in \mathcal{P}$ and $\bar{Y}(p^\star + td) \geq \bar{Y}(p^\star) + \kappa t$, for $t \in [0, t_0]$. Then, as $\epsilon \to 0$,*

$$V^\star_{\epsilon,G_0} - R^\star_{G_0} = O(\epsilon).$$

Theorem 2 relies on two local properties near the optimal policy $p^\star$: the average treatment probability must be Lipschitz continuous, and there must exist a direction of growth that allows the average outcome to strictly increase. These conditions ensure that a strictly feasible policy can be constructed to satisfy the surrogate's ϵ-buffer without a disproportionate jump in resource costs. Provided that $R_N(\mathcal{P}) = O(N^{-1/2})$, the regret bound is $O(\epsilon + \epsilon^{-1}N^{-1/2})$. We can then choose the rate of ϵ optimally by setting $\epsilon \propto N^{-1/4}$, which leads to a regret bound of order $O(N^{-1/4})$.

4.2 Regret Rate Refinement

The previous result yields a convergence rate of $N^{-1/4}$, which reflects the approximation–estimation tradeoff induced by the surrogate risk, similar to that arising in surrogate-based policy learning (e.g., Chen et al., 2016). However, this rate can be improved by exploiting the zero-penalty region of the surrogate loss, provided that the outcome target is attainable with positive slack. Specifically, define the maximum attainable slack as

$$s := \sup_{p\in\mathcal{P}} \bar{Y}(p) - T, \tag{11}$$

[3] By the Contraction Lemma (Shalev-Shwartz and Ben-David, 2014, Lemma 26.9), the result follows since the logistic function is 1/4-Lipschitz and the logistic policy class is obtained by composing the linear class with $\sigma(\cdot)$.

and, for $\delta \in (0,1)$, let

$$B_{\text{I-IPW}}(\delta) = \frac{M}{\eta}(2n_{\max})^{3/2}\mathcal{R}_N(\mathcal{P}) + \frac{M}{\eta}\sqrt{\frac{2\log(2/\delta)}{N}}.$$

By Lemma C.2, $B_{\text{I-IPW}}(\delta)$ bounds the uniform IPW outcome estimation error with probability at least $1-\delta$. Further, let $c = c(C_0) \in (0,1)$ denote the level at which $h_\epsilon(T + c\,\epsilon; T) = 1/C_0$:

$$c(C_0) = \begin{cases} \sqrt{(1-1/C_0)/2}, & 1 < C_0 \le 2, \\ 1 - (2C_0)^{-1/2}, & C_0 \ge 2, \end{cases}$$

Theorem 3 (Expected regret refinement)**.** *Suppose Assumptions 1–3 hold, and that $s > 0$. For any $\delta \in (0,1)$ and $\epsilon > 0$ satisfying*

$$B_{\text{I-IPW}}(\delta) \le c\epsilon, \qquad \epsilon + B_{\text{I-IPW}}(\delta) < s,$$

the IPW policy estimator satisfies

$$\text{Reg}(\hat{p}_{\text{I-IPW}}) \le \underbrace{\bar{C}^\star\big(T + \epsilon + B_{\text{I-IPW}}(\delta)\big) - \bar{C}^\star(T)}_{(d)} + \underbrace{4n_{\max}\mathcal{R}_N(\mathcal{P})}_{(e)} + \underbrace{(1+C_0)\delta}_{(f)}.$$

The regret bound of Theorem 3 has a similar structure to Theorem 1, and depends on three terms. Term (d) in the regret bound represents the additional treatment "cost" induced by using the surrogate risk with the calibrated buffer $\epsilon \ge B_{\text{I-IPW}}(\delta)/c$, and can be interpreted as an approximation error, similarly to term (a) in Theorem 1. Term (e) controls estimation error in the average treatment probability, and term (f) bounds the constraint violation probability. The bound thus reflects a related trade-off to that of Theorem 1 between the cost of ensuring feasibility and the confidence level of that guarantee: decreasing δ reduces term (f), but increases $B_{\text{I-IPW}}(\delta)$, requiring a larger ϵ and potentially increasing term (d).

Theorem 3 obtains the regret bound by exploiting the zero-penalty region of the surrogate loss, thereby avoiding the $1/\epsilon$ factor arising from uniform control of the surrogate risk. The condition $\epsilon + B_{\text{I-IPW}}(\delta) < s$ ensures that the buffered target is attainable. On the event that the uniform IPW outcome estimation error is bounded by $B_{\text{I-IPW}}(\delta)$, which holds with probability at least $1-\delta$, policies attaining this buffered target incur zero empirical penalty. Lemma C.4 then implies that $\widehat{\bar{Y}}^{\text{I-IPW}}(\hat{p}_{\text{I-IPW}}) \ge T + c\epsilon$, so $\epsilon \ge B_{\text{I-IPW}}(\delta)/c$ guarantees population feasibility. The regret bound follows by comparing treatment costs with those of policies attaining the buffered target $T + \epsilon + B_{\text{I-IPW}}(\delta)$ (Appendix C.4). Such regret bound yields a sharper convergence rate. In particular, let $\delta_N = N^{-1}$ and $\epsilon_N = B_{\text{I-IPW}}(\delta_N)/c$. If $\mathcal{R}_N(\mathcal{P}) = O(N^{-1/2})$, then $\epsilon_N + B_{\text{I-IPW}}(\delta_N) = O(\sqrt{\log N/N})$, so the conditions of Theorem 3 hold for all $N \ge N_0$, for some N_0. Under the additional conditions of Theorem 2, the

approximation error in the average treatment probability is also $O(\sqrt{\log N/N})$, yielding

$$\mathrm{Reg}(\widehat{p}_{\text{I-IPW}}) = O\left(\sqrt{\frac{\log N}{N}}\right).$$

This suggests choosing the margin ϵ_N large enough to dominate the uniform bound $c^{-1}B_{\text{I-IPW}}(\delta_N)$. In practice, however, the constants entering $B_{\text{I-IPW}}(\delta_N)$ may be difficult to obtain or may result in overly conservative margins, producing policies with high treatment probabilities.

Cross-validated margin. As a practical alternative, we propose a cross-validation procedure to calibrate the finite-sample constant in the margin. Let $\epsilon_N(c) = c\sqrt{\log N/N}$, where $c \in \mathcal{C} = \{c_1, \ldots, c_L\}$ is selected from a prespecified grid. We use a cluster-level K-fold cross-validation to select c, as summarized in Algorithm 1.

Algorithm 1 Cross-validated selection of the IPW margin

Require: Data from N clusters, target T, grid $\mathcal{C}$, and K folds.

1: Partition the clusters into folds $\mathcal{I}_1, \ldots, \mathcal{I}_K$.

2: **for** $c \in \mathcal{C}$ and $k = 1, \ldots, K$ **do**

3: Fit on $\mathcal{I}_{-k}$, containing N_{-k} clusters,

$$\widehat{p}_c^{(-k)} \in \arg\min_{p \in \mathcal{P}} \left\{ \widehat{C}_{-k}(p) + C_0 h_{c/\sqrt{N_{-k}}}\big(\widehat{Y}_{-k}^{\text{I-IPW}}(p); T\big) \right\}.$$

4: On fold $\mathcal{I}_k$, compute

$$Y_{k,c} = \widehat{Y}_k^{\text{I-IPW}}(\widehat{p}_c^{(-k)}), \qquad C_{k,c} = \widehat{C}_k(\widehat{p}_c^{(-k)}).$$

5: **end for**

6: For each c, calculate

$$F(c) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\{Y_{k,c} \geq T\}, \quad S(c) = \frac{1}{K}\sum_{k=1}^{K}(T - Y_{k,c})_+, \quad R(c) = \frac{1}{K}\sum_{k=1}^{K} C_{k,c}.$$

7: Select

$$\widehat{c}_N = \operatorname{lex}\arg\min_{c \in \mathcal{C}} \big(-F(c), S(c), R(c), c\big).$$

8: **return** $\widehat{c}_N$.

Here, lex arg min denotes lexicographic minimization, and $(x)_+ = \max\{x, 0\}$. For each candidate c, the policy is fitted on $K-1$ folds and evaluated on the held-out fold. Candidates are ranked lexicographically, prioritizing the proportion of folds satisfying the outcome target, followed by the average magnitude of constraint violations and the average treatment probability. Remaining ties are broken in favor of the smaller c. The selected constant is then used to refit the policy on the full sample[4].

[4]Cross-validation and holdout evaluation are standard tools in policy learning for choosing tuning pa-

In the simulation study of Section 7, our cross-validated margin shrinks with the sample size N and the resulting policies attain high feasibility rates with regret decreasing in N. In the empirical application of Section 8, the estimated policies using this procedure are feasible in most held-out folds or fall short of the target only marginally.

4.3 Regret Bound for the Hájek Estimator

We next establish uniform finite-sample deviation bounds for the self-normalized Hájek estimator over potentially infinite policy classes and use them to show that Hájek-based policy learning preserves the regret rates established for IPW. We restrict attention to policies satisfying $0 < p(\boldsymbol{x}, \boldsymbol{v}) < 1$, which ensures that the Hájek denominator $D_N(p)$ is positive for every policy and sample.[5] Analogously to $B_{\text{I-IPW}}(\delta)$ in Section 4.2, define, for $\delta \in (0,1)$, the uniform deviation bound

$$B_{\text{I-HJ}}(\delta) = 6\sqrt{2}\frac{M}{\eta} n_{\max}^{3/2} \mathcal{R}_N(\mathcal{P}) + \frac{2M}{\eta\sqrt{N}} + \frac{M}{\eta}\sqrt{\frac{2\log(2/\delta)}{N}}. \tag{12}$$

Lemma C.3 then gives $\sup_{p\in\mathcal{P}} |\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)| \le B_{\text{I-HJ}}(\delta)$ with probability at least $1-\delta$.

Theorem 4 (Regret under Hájek normalization)**.** *Suppose Assumptions 1–3 hold,* $0 < p(\boldsymbol{x}, \boldsymbol{v}) < 1$ *for every* $p \in \mathcal{P}$ *and* $(\boldsymbol{x}, \boldsymbol{v}) \in \mathcal{X} \times \mathcal{V}$.

1. *For any* $\epsilon > 0$,

$$\text{Reg}(\widehat{p}_{\text{I-HJ}}) \le V^{\star}_{\epsilon, G_0} - R^{\star}_{G_0} + 8n_{\max} R_N(\mathcal{P}) + 24\sqrt{2}\,\frac{C_0 M}{\eta\epsilon} n_{\max}^{3/2} \mathcal{R}_N(\mathcal{P}) + \frac{4C_0 M}{\eta\epsilon\sqrt{N}}\left(2 + \sqrt{2\log 2}\right).$$

2. *Suppose additionally that* $s > 0$. *For any* $\delta \in (0,1)$ *and* $\epsilon > 0$ *satisfying* $c\epsilon \ge B_{\text{I-HJ}}(\delta)$. *and* $\epsilon + B_{\text{I-HJ}}(\delta) < s$ *it holds:*

$$\text{Reg}(\widehat{p}_{\text{I-HJ}}) \le \bar{C}^{\star}(T + \epsilon + B_{\text{I-HJ}}(\delta)) - \bar{C}^{\star}(T) + 8n_{\max}\mathcal{R}_N(\mathcal{P}) + (1 + C_0)\delta.$$

Theorem 4 shows that Hájek normalization preserves the regret rates obtained with the IPW constraint estimator. In particular, if $\mathcal{R}_N(\mathcal{P}) = O(N^{-1/2})$ and the approximation error satisfies the conditions of Theorem 2, part (i) yields regret $O(\epsilon + \epsilon^{-1}N^{-1/2})$, giving $O(N^{-1/4})$ for $\epsilon \asymp N^{-1/4}$. Part (ii) yields the sharper rate $O(\sqrt{\log N/N})$ by taking $\delta_N = N^{-1}$ and $\epsilon_N = B_{\text{I-HJ}}(\delta_N)/c$.

rameters based on estimated out-of-sample performance (Zhao et al., 2012; Mbakop and Tabord-Meehan, 2021). Alternatively, $B_{\text{I-IPW}}(\delta)$ could be calibrated by a cluster-level multiplier bootstrap approximation to quantiles of the uniform outcome estimation error over the policy class (e.g. Kennedy, 2019), albeit at a high computational cost. We leave this possibility for future research.

[5] Restriction to this policy classes is imposed to avoid defining the Hájek estimator on zero-denominator samples, and is made for analytical convenience. Policy classes containing 0 and 1 can be accommodated by setting $\widehat{\bar{Y}}_{\text{I-HJ}}(p) = 0$ when $D_N(p) = 0$, since the centered-moment identity underlying the deviation bounds remain valid under this convention (see Lemma C.3).

The rate equivalence in Theorem 4 is expected, paralleling the well-known consistency rate equivalence between Hájek and IPW estimators for average treatment effects with and without interference (Hájek, 1971; Aronow and Samii, 2017). Nevertheless, Hájek normalization can reduce variance and improve finite-sample stability, an advantage not captured by the regret bound rates. Consistent with this, in our simulations under uniform margins, Hájek-based policies lead to a lower average treatment probability while achieving comparable or higher feasibility and lower regret than IPW-based policies at every sample size. These gains are not universal, however, as the Hájek approach does not uniformly outperform IPW in our empirical application. Similar finite-sample advantages of self-normalization in policy learning without interference have been studied, for instance, by Khan and Ugander (2023) and Swaminathan and Joachims (2015b).

5 Variance-Adaptive Margin

In the previous sections, the margin parameter ϵ_N was chosen uniformly over all policies in $\mathcal{P}$, ignoring the heterogeneous variability of their outcome estimates. This uniform approach introduces two sources of inefficiency. First, it over-penalizes low-variance policies whose true performance lies close to the threshold T but can be estimated with high precision. Second, it fails to adequately guard against high-variance policies, which require wider margins to protect against apparent feasibility driven by sampling noise. By ignoring these differences in estimation uncertainty, a uniform margin may lead to unnecessarily high treatment probabilities in finite samples. To address this issue, we consider a policy-dependent margin that adapts to the estimated variability of the outcome under each candidate policy. For the IPW constraint, the relevant quantity is the sample variance of $\psi_i^{\text{I-IPW}}(p)$, defined in (8). Then for each $p \in \mathcal{P}$, we set

$$\widehat{\epsilon}_N(p) = \lambda_N \sqrt{\frac{\widehat{\text{Var}}\{\psi_i^{\text{I-IPW}}(p)\}}{N}} + r_N,$$

where λ_N controls the degree of conservativeness and $r_N \geq 0$ is a lower-order additive correction. The variance-dependent component assigns smaller margins to policies with more precise outcome estimates and larger margins to those with greater uncertainty, potentially reducing resource use while maintaining protection against constraint violations.

This construction is motivated by empirical Bernstein-type generalization bounds (Maurer and Pontil, 2009; Swaminathan and Joachims, 2015a), which replace worst-case variance bounds with policy-specific empirical variances. The next theorem establishes regret guarantees for theoretically calibrated λ_N and r_N. In practice, we use the leading variance-dependent term, setting $r_N = 0$ and selecting λ_N by cross-validation.

Theorem 5 (Variance-adaptive margin)**.** *Suppose Assumptions 1–3 hold and $s > 0$. For any $N \geq 16$ and $\delta \in (0,1)$, let $\Lambda_N := \Lambda_{N,\text{I-IPW}}(\delta/2)$, defined in Theorem C.1 of Appendix C.1.*

Set $\lambda_N = \sqrt{18\Lambda_N}/c$ and $r_N = 32M\Lambda_N/(c\eta N)$, and for each $p \in \mathcal{P}$ define the empirical margin

$$\widehat{\epsilon}_N(p) = \lambda_N \sqrt{\frac{\widehat{\sigma}^2_{\text{I-IPW}}(p)}{N}} + r_N = \frac{1}{c}\left\{\sqrt{\frac{18\,\widehat{\sigma}^2_{\text{I-IPW}}(p)\,\Lambda_N}{N}} + \frac{32M\Lambda_N}{\eta N}\right\}.$$

Let

$$\widehat{p}^{\,\text{VA}}_{\text{I-IPW}} \in \arg\min_{p\in\mathcal{P}}\Big\{\widehat{\bar{C}}(p) + C_0\, h_{\widehat{\epsilon}_N(p)}\big(\widehat{\bar{Y}}_{\text{I-IPW}}(p); T\big)\Big\}.$$

Define the population buffer and the corresponding set of buffered-feasible policies,

$$\Delta_N(p) = d_1\,\sigma_{\text{I-IPW}}(p)\sqrt{\frac{\Lambda_N}{N}} + d_2\,\frac{M}{\eta}\frac{\Lambda_N}{N}, \qquad \mathcal{A}^{\Delta}_N = \big\{p \in \mathcal{P} : \bar{Y}(p) \geq T + \Delta_N(p)\big\},$$

where $d_1 \geq (1{+}c^{-1})\sqrt{18}$ and $d_2 \geq (1{+}c^{-1})(16\sqrt{18}{+}32)$ depend only on C_0. If $\sup_{p\in\mathcal{P}} \Delta_N(p) < s$, then

$$\text{Reg}\big(\widehat{p}^{\,\text{VA}}_{\text{I-IPW}}\big) \leq \underbrace{\inf_{p\in\mathcal{A}^{\Delta}_N} \bar{C}(p) - \bar{C}^{\star}(T)}_{(d')} + \underbrace{4n_{\max}\mathcal{R}_N(\mathcal{P})}_{(e')} + \underbrace{(1+C_0)\delta}_{(f')}.$$

The quantities in Theorem 5 have the following interpretation. $\sigma^2_{\text{I-IPW}}(p) = \text{Var}_{G_0}\{\psi_i^{\text{I-IPW}}(p)\}$ and $\widehat{\sigma}^2_{\text{I-IPW}}(p)$ denote the population and sample IPW variances. The term Λ_N captures the complexity of $\mathcal{P}$ through its sup-norm covering numbers $N_\infty(r, \mathcal{P}, m)$, which measure the number of functions needed to approximate the policy class uniformly over m evaluation points (see Appendix C.1). The theoretical calibration ensures that $c\widehat{\epsilon}_N(p)$ is a uniform empirical Bernstein confidence radius, satisfying $|\widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p)| \leq c\widehat{\epsilon}_N(p)$ simultaneously for all $p \in \mathcal{P}$ with probability at least $1 - \delta/2$ (Theorem C.1 in Appendix C.1). The population buffer $\Delta_N(p)$ is constructed to dominate $(1+c)\widehat{\epsilon}_N(p)$ uniformly with high probability, ensuring that policies satisfying $\bar{Y}(p) \geq T + \Delta_N(p)$ incur zero empirical surrogate penalty. Consequently, term (d') represents the additional treatment cost induced by the population buffer $\Delta_N(p)$ associated with the policy-specific margin $\widehat{\epsilon}_N(p)$, analogous to the approximation term in Theorem 3 for a uniform margin. Terms (e') and (f') are the same as terms (e) and (f) of Theorem 3

To obtain the convergence rate, let $\delta_N = N^{-1}$. For linear or logistic policy classes with bounded covariates and a fixed-dimensional bounded parameter space, $\Lambda_{N,\text{I-IPW}}(\delta_N/2) = O(\log N)$, so $\sup_{p\in\mathcal{P}} \Delta_N(p) = O(\sqrt{\log N/N})$[6]. Hence, under positive slack, $\mathcal{A}^{\Delta}_N$ is nonempty for all sufficiently large N. Combined with $\mathcal{R}_N(\mathcal{P}) = O(N^{-1/2})$ and the conditions of

[6]This follows, for instance, by combining Wainwright (2019), Lemma 5.7, with the covering-number inequality (C.4) in Theorem C.1 of Appendix C.1.

Theorem 2, this yields

$$\mathrm{Reg}(\widehat{p}^{\mathrm{VA}}_{\text{I-IPW}}) = O\left(\sqrt{\frac{\log N}{N}}\right).$$

Although the variance-adaptive procedure achieves the same regret rate as its uniform-margin counterpart, its policy-specific margin can reduce conservativeness and improve finite-sample stability by allowing policies with lower estimated outcome variance to satisfy the constraint with smaller margins. In particular, if $\Delta_N(p) \leq \epsilon_N + B_{\text{I-IPW}}(\delta_N)$ for all $p \in \mathcal{P}$, the set $\Delta_N(p)$ contains the uniform-margin one, yielding a weakly smaller approximation term (d') compared to (d) in Theorem 3. While these potential advantages are not captured by the common convergence rate, our simulations show that variance-adaptive IPW achieves lower and less variable treatment probabilities and lower regret at every N, with comparable feasibility. In the Colombian application, variance-adaptive IPW similarly reduces average treatment probability across outcome targets.

The variance-adaptive construction can be extended to the Hájek estimator. For a policy p, define

$$\widehat{\sigma}^2_{\text{I-HJ}}(p) = \frac{1}{N-1}\sum_{i=1}^{N}\left\{\widehat{\phi}^{\text{I-HJ}}_i(p)\right\}^2, \quad \widehat{\phi}^{\text{I-HJ}}_i(p) = \frac{w_i(p)\{\bar{Y}_i - \widehat{\bar{Y}}_{\text{I-HJ}}(p)\}}{D_N(p)},$$

We set the policy-specific margin to $\widehat{\epsilon}_{N,\text{I-HJ}}(p) = \lambda_N \frac{\widehat{\sigma}_{\text{I-HJ}}(p)}{\sqrt{N}}$, and define the following Hájek policy estimator:

$$\hat{p}^{\mathrm{VA}}_{\text{I-HJ}} \in \arg\min_{p\in\mathcal{P}} \widehat{V}^{VA}_{\text{I-HJ}}(p), \quad \text{with} \quad \widehat{V}^{VA}_{\text{I-HJ}}(p) = \widehat{\bar{C}}(p) + C_0 h_{\widehat{\epsilon}_{N,\text{I-HJ}}(p)}\left(\widehat{\bar{Y}}_{\text{I-HJ}}(p); T\right).$$

The relevant population variance differs from that of the unnormalized IPW estimator because self-normalization incorporates the covariance between the weighted outcome and the weights. Consequently, the Hájek margin can be smaller for policies for which normalization reduces estimation variance, although no uniform ordering between the two estimators is implied. We establish a uniform empirical Bernstein bound for the Hájek estimator in Appendix C.1 (Theorem C.2), which yields a corresponding variance-adaptive regret guarantee for $\hat{p}^{\mathrm{VA}}_{\text{I-HJ}}$ using a theoretically calibrated margin (Theorem C.3, Appendix C.7). The resulting regret bound controls excess resource use relative to policies satisfying a variance-dependent population buffer determined by the Hájek estimator variance (see Eq. (C.16) in Theorem C.3, Appendix C.7).

6 Resource-Constrained Problem

A complementary problem maximizes the population average outcome subject to a constraint on the average treatment probability:

$$p^\star \in \arg\max_{p \in \mathcal{P}} \bar{Y}(p) \quad \text{subject to} \quad \bar{C}(p) \leq B, \tag{13}$$

where B is a prespecified resource budget. Appendix A.1 studies this resource-constrained formulation and establishes equivalence to the outcome-target problem at appropriately matched thresholds under a condition of local outcome growth. This condition implies that any policy leaving resources unused admits an arbitrarily small adjustment that strictly improves the average outcome–a property that holds, for instance, when the average outcome is strictly increasing in treatment.

The two formulations differ in the statistical complexity of their constraints. The resource constraint depends only on observed covariates and policy probabilities, so it is simple to estimate and feasibility may be easier to attain. By contrast, the outcome constraint in (3) requires interference-aware estimation of counterfactual outcomes, so ignoring interference can distort the estimated feasible set. In our simulations under the same data-generating process in Appendix A.2, the resource-constrained estimators achieve uniformly high feasibility and the no-interference estimator performs comparably to interference-aware methods, whereas in the outcome-target problem the no-interference estimator selects infeasible policies. One might think that the resource-constrained problem is easier to learn than the outcome-target problem, since its constraint requires no counterfactual estimation and feasibility is therefore easier to guarantee. However, its objective still requires interference-aware outcome estimation, so its regret guarantees still depend on outcome estimation error. The simpler estimation of the resource constraint and the equivalence of the two problems at appropriately matched thresholds also suggest that a potential alternative for learning an outcome-target policy is to solve the resource-constrained problem over a grid of budget levels and invert the estimated welfare frontier to identify the minimum B attaining T. However, determining this budget level by inverting the estimated welfare frontier remains sensitive to outcome estimation uncertainty. Developing such frontier-inversion procedures and comparing their performance with direct outcome-target learning is an interesting direction for future research.

7 Simulation Study

We simulate 500 datasets for each $N \in \{200, 400, 800, 2000\}$, with clusters of size $n_i = 10$. For each individual, we generate a single binary covariate $X_{ij} \sim$ Bernoulli(0.6) and

independently assign treatment as $A_{ij} \sim \text{Bernoulli}(0.5)$. Let $G_{1ij} = (n_i - 1)^{-1} \sum_{k \neq j} X_{ik} A_{ik}$ and $G_{0ij} = (n_i - 1)^{-1} \sum_{k \neq j} (1 - X_{ik}) A_{ik}$ denote the proportions of treated peers with $X = 1$ and $X = 0$, respectively. Then, the outcome is generated by

$$Y_{ij} = \beta_0 + \kappa_0 A_{ij}(1 - X_{ij}) + \kappa_1 A_{ij} X_{ij} + \theta_1 G_{1ij}(1 - A_{ij}) X_{ij} + \theta_0 G_{0ij}(1 - A_{ij})(1 - X_{ij}) + u_{ij}, \quad (14)$$

where $u_{ij} \sim \mathcal{N}(0, 1)$. Note that the terms involving G_{1ij} and G_{0ij} capture heterogeneous spillover exposure within clusters. We set $\beta_0 = 0.15$, $\kappa_0 = 2.25$, $\kappa_1 = 1.00$, $\theta_0 = 0.30$, $\theta_1 = 0.70$, and the outcome target $T = 0.75$. We use the logistic class $\mathcal{P}_{\Gamma,\text{logit}}$ of Section 2.3, with $p_\gamma(X_{ij}) = \sigma(\gamma_0 + \gamma_1 X_{ij})$ and $\Gamma = [-1, 1]^2$, and denote by $\mathcal{P}_{\Gamma_0,\text{logit}}$, with $\Gamma_0 = [-1, 1] \times \{0\}$, its constant subclass.

First, we evaluate four interference-aware estimators combining IPW or Hájek outcome estimation with uniform (U) or variance-adaptive (VA) margins: I-IPW-U, I-IPW-VA, I-HJ-U, and I-HJ-VA. Margin constants are selected in each replication by 5-fold cluster-level cross-validation (Algorithm 1). Second, we compare the proposed interference-aware estimators with two benchmarks. The first is a no-interference estimator, NI-IPW, which uses the same individualized logistic policy class but replaces the cluster-level IPW estimator with the following IPW estimator ignoring interference

$$\hat{\bar{Y}}_{\text{NI-IPW}}(p) = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij} \frac{\mathbb{P}_p(A_{ij} \mid \boldsymbol{X}_{ij}, \boldsymbol{V}_i; p)}{\mathbb{P}_{G_0}(A_{ij} \mid \boldsymbol{X}_i, \boldsymbol{V}_i)}.$$

The second benchmark, the constant-policy estimator CI-IPW, retains the interference-aware IPW estimator but restricts optimization to the constant subclass $\mathcal{P}_{\Gamma_0,\text{logit}}$. NI-IPW isolates the consequences of ignoring interference, while CI-IPW assesses the gains from personalizing treatment probabilities. As population benchmarks, we report the population risk $R_{G_0}(p)$, average treatment probability $\bar{C}(p)$, and average outcome $\bar{Y}(p)$ evaluated at the following policies: (i) the optimal personalized policy $p^\star \in \arg\min_{p \in \mathcal{P}_{\Gamma,\text{logit}}} R_{G_0}(p)$; (ii) the optimal constant policy $q^\star \in \arg\min_{p \in \mathcal{P}_{\Gamma_0,\text{logit}}} R_{G_0}(p)$; and (iii) the constant policy $\tilde{q} = \bar{C}(p^\star)$, which treats every unit uniformly with probability equal to $\bar{C}(p^\star)$.

Table 1 summarizes the performance of the four interference-aware procedures. The optimal personalized policy attains the target with an average treatment probability of 0.318. For all four procedures, the regret and constraint margins generally decrease with N, while feasibility rates remain high (0.984–0.998). I-IPW-U has the highest and most variable average treatment probabilities. For instance, at $N = 200$ it attains the highest mean outcome (0.965) but lowest feasibility (0.984), with the highest average treatment probability (0.461). Relative to I-IPW-U, I-IPW-VA uses smaller margins and lower treatment probabilities, with comparable feasibility and lower regret at every N. Under uniform margins, Hájek

Table 1: Monte Carlo performance of the four interference-aware procedures.

N	Estimator	Regret	Treat. Probability	Outcome	Feasibility	Margin
200	I-IPW-U	0.160 (0.119)	0.461 (0.070)	0.965 (0.105)	0.984 (0.126)	0.2412
	I-IPW-VA	0.100 (0.066)	0.414 (0.027)	0.857 (0.032)	0.996 (0.063)	0.1214
	I-HJ-U	0.125 (0.072)	0.442 (0.061)	0.950 (0.094)	0.998 (0.045)	0.2052
	I-HJ-VA	0.104 (0.067)	0.418 (0.035)	0.900 (0.055)	0.996 (0.063)	0.1596
400	I-IPW-U	0.108 (0.082)	0.420 (0.047)	0.913 (0.074)	0.994 (0.077)	0.1706
	I-IPW-VA	0.084 (0.078)	0.396 (0.024)	0.839 (0.026)	0.994 (0.077)	0.0994
	I-HJ-U	0.091 (0.083)	0.403 (0.038)	0.891 (0.063)	0.994 (0.077)	0.1431
	I-HJ-VA	0.083 (0.080)	0.396 (0.029)	0.869 (0.045)	0.994 (0.077)	0.1242
800	I-IPW-U	0.088 (0.115)	0.392 (0.034)	0.872 (0.056)	0.986 (0.118)	0.1276
	I-IPW-VA	0.080 (0.126)	0.381 (0.023)	0.826 (0.024)	0.984 (0.126)	0.0860
	I-HJ-U	0.075 (0.110)	0.381 (0.029)	0.855 (0.047)	0.988 (0.109)	0.1079
	I-HJ-VA	0.070 (0.101)	0.378 (0.025)	0.844 (0.039)	0.990 (0.100)	0.0989
2000	I-IPW-U	0.058 (0.091)	0.368 (0.022)	0.834 (0.036)	0.992 (0.089)	0.0828
	I-IPW-VA	0.055 (0.090)	0.365 (0.018)	0.815 (0.021)	0.992 (0.089)	0.0672
	I-HJ-U	0.045 (0.049)	0.362 (0.019)	0.823 (0.032)	0.998 (0.045)	0.0720
	I-HJ-VA	0.046 (0.066)	0.361 (0.018)	0.820 (0.029)	0.996 (0.063)	0.0696

Notes: Monte Carlo means (standard deviations) over 500 replications. Treat. probability is $\bar{C}(\hat{p})$; Outcome is $\bar{Y}(\hat{p})$; Feasibility is $1\{\bar{Y}(\hat{p}) \geq T\}$. Regret refers to $R_{G_0}(\hat{p}) - R^{\star}_{G_0}$, where $R(p) = \bar{C}(p) + 1.05\, 1\{\bar{Y}(p) < T\}$, $R^{\star}_{G_0} = 0.3185$, and $T = 0.75$.

normalization similarly reduces the treatment probability and regret at every N, with feasibility at least as high. The gains from variance-adaptive margins are smaller for the Hájek estimator and vanish at $N = 2000$. Overall, the proposed procedures perform well, with variance-adaptive margins yielding lower average treatment probabilities with comparable feasibility, particularly when combined with IPW estimation.

Table 2 compares the interference-aware estimators with NI-IPW and CI-IPW, using uniform margins throughout for brevity. Ignoring interference leads to systematically infeasible policies: NI-IPW assigns lower treatment probabilities, but its feasibility rate falls from 0.874 at $N = 200$ to 0.142 at $N = 2000$, where its mean population outcome is 0.729, below target. This occurs because the no-interference estimator evaluates spillover exposure at the experimental treatment rate rather than under the candidate policy, systematically overestimating outcomes for policies with lower treatment probabilities in our design. In contrast, CI-IPW achieves feasibility comparable to I-IPW. In small samples it also attains lower regret, but for $N \geq 800$ it requires a higher treatment probability: 0.383 at $N = 2000$, compared with 0.368 for I-IPW. The constant-policy estimator also exhibits less variable average treatment probability and smaller margins, reflecting the lower complexity of the constant policy class. The population benchmarks isolate the gain from assigning treatment probabilities based on covariates through two complementary comparisons. The optimal personalized policy $p^{\star}$ attains the target with $\bar{C}(p^{\star}) = 0.3185$, whereas the constant policy $\tilde{q}$, which treats every unit with this same probability and thus has $\bar{C}(\tilde{q}) = 0.3185$, achieves only

Table 2: Monte Carlo performance of the uniform-margin procedures and benchmarks.

N	Estimator	Regret	Treat. Probability	Outcome	Feasibility	Margin
200	I-IPW	0.160 (0.119)	0.461 (0.070)	0.965 (0.105)	0.984 (0.126)	0.2412
	I-HJ	0.125 (0.072)	0.442 (0.061)	0.950 (0.094)	0.998 (0.045)	0.2052
	CI-IPW	0.112 (0.027)	0.430 (0.027)	0.869 (0.041)	1.000 (0.000)	0.1154
	NI-IPW	0.170 (0.331)	0.356 (0.038)	0.813 (0.063)	0.874 (0.332)	0.1073
400	I-IPW	0.108 (0.082)	0.420 (0.047)	0.913 (0.074)	0.994 (0.077)	0.1706
	I-HJ	0.091 (0.083)	0.403 (0.038)	0.891 (0.063)	0.994 (0.077)	0.1431
	CI-IPW	0.093 (0.048)	0.410 (0.019)	0.837 (0.030)	0.998 (0.045)	0.0846
	NI-IPW	0.289 (0.444)	0.334 (0.024)	0.776 (0.041)	0.740 (0.439)	0.0755
800	I-IPW	0.088 (0.115)	0.392 (0.034)	0.872 (0.056)	0.986 (0.118)	0.1276
	I-HJ	0.075 (0.110)	0.381 (0.029)	0.855 (0.047)	0.988 (0.109)	0.1079
	CI-IPW	0.087 (0.090)	0.397 (0.017)	0.817 (0.026)	0.992 (0.089)	0.0671
	NI-IPW	0.530 (0.512)	0.319 (0.017)	0.751 (0.030)	0.496 (0.500)	0.0544
2000	I-IPW	0.058 (0.091)	0.368 (0.022)	0.834 (0.036)	0.992 (0.089)	0.0828
	I-HJ	0.045 (0.049)	0.362 (0.019)	0.823 (0.032)	0.998 (0.045)	0.0720
	CI-IPW	0.075 (0.102)	0.383 (0.011)	0.795 (0.018)	0.990 (0.100)	0.0437
	NI-IPW	0.889 (0.359)	0.306 (0.011)	0.729 (0.020)	0.142 (0.349)	0.0339
Population benchmarks:						
Optimal personalized policy	$p^\star$:	$R_{G_0}(p^\star) = 0.3185$,	$\bar{C}(p^\star) = 0.3185$,	$\bar{Y}(p^\star) = 0.750$		
Optimal constant policy	$q^\star$:	$R_{G_0}(q^\star) = 0.354$,	$\bar{C}(q^\star) = 0.354$,	$\bar{Y}(q^\star) = 0.750$		
Constant policy	$\tilde{q}$:	$R_{G_0}(\tilde{q}) = 1.3685$,	$\bar{C}(\tilde{q}) = 0.3185$,	$\bar{Y}(\tilde{q}) = 0.693$		

Notes: Monte Carlo means (standard deviations) over 500 replications. Treat. probability is $\bar{C}(\hat{p})$; Outcome is $\bar{Y}(\hat{p})$; Feasibility is $1\{\bar{Y}(\hat{p}) \geq T\}$. Regret refers to $R_{G_0}(\hat{p}) - R^\star_{G_0}$, where $R(p) = \bar{C}(p) + 1.05\, 1\{\bar{Y}(p) < T\}$, $R^\star_{G_0} = 0.3185$, and $T = 0.75$. Population benchmarks are evaluated exactly under the data-generating process: $p^\star$ is the optimal policy in $\mathcal{P}_{\Gamma,\text{logit}}$; $q^\star$ is the optimal constant policy in $\mathcal{P}_{\Gamma_0,\text{logit}}$; and $\tilde{q} = \bar{C}(p^\star) = 0.3185$ is the constant policy matching the average treatment probability attained at $p^\star$.

$\bar{Y}(\tilde{q}) = 0.693 < T$. The optimal constant policy achieves the outcome target T, however it yields $\bar{C}(q^\star) = 0.354$, 3.6 percentage points more than under $p^\star$.

8 Empirical Application

We apply our methodology to the 2005 randomized experiment evaluating the *Subsidios Condicionados a la Asistencia Escolar*, a conditional cash transfer program promoting school attendance and post-secondary enrollment among low-income students in Bogotá, Colombia (Barrera-Osorio et al., 2011). Students from households with one to five school-aged children were individually randomized within strata defined by locality, school type, gender, and grade. Transfers were conditional on maintaining at least 80 percent monthly school attendance. This setting creates scope for within-household spillovers. Transfers may encourage siblings' attendance through shared academic environments, peer influence, or parental awareness, but may also divert parental resources toward treated children, reducing attendance among untreated siblings (Camilo and Zuluaga, 2022; Ferreira et al., 2009). Additionally, both direct and spillover effects may be heterogeneous. Barrera-Osorio et al.

(2011) document heterogeneity in the direct response by gender and economic status, and, notably, find that adverse spillovers are asymmetric by gender: untreated female siblings of a treated student experience a sharper attendance decline than untreated male siblings. Spillovers may further depend on household size, since the number of eligible children affects competition for household educational resources (Park and Kang, 2022).

Motivated by this evidence, we estimate an optimal policy that attains a prespecified aggregate attendance target while accounting for heterogeneous direct and spillover effects. Let $Y_{ij} \in [0, 1]$ denote student j's monitored attendance rate in household i, and $A_{ij} \in \{0, 1\}$ indicate assignment to either cash-transfer intervention. We specify a logistic policy using four covariates: gender, to capture heterogeneous direct and spillover effects; SISBEN socioeconomic status, to capture financial constraints and baseline dropout risk; household size, to capture within-household interference; and locality, to account for differences between experimental sites. Restricting to complete cases yields 6,729 students in 5,725 analytic clusters. We compare I-IPW, I-HJ, NI-IPW, and CI-IPW policies to assess the consequences of ignoring interference and the benefits of personalization.

Out-of-sample evaluation We conduct five-fold cross-validation at the household level. We randomly split the households into five folds and, for each fold, estimate the policy on the remaining folds and evaluate it on the held-out fold. Table 3 reports, averaged over folds, the average treatment probability of each learned policy (*Avg. prob.*), its interference-aware IPW outcome (*IPW outcome*), the share of folds in which it attains the target (*Feasibility*), and its average shortfall below the target (*Welfare loss*). For comparison, *Own outcome* reports the outcome under each procedure's own estimator.

At lower targets, accounting for interference and personalizing treatment probabilities yield substantial resource savings. At $T = 0.77$, I-IPW reduces treatment by 9.2 and 22.3 percentage points relative to NI-IPW and CI-IPW, respectively, with savings of approximately 10 percentage points at $T = 0.78$. At the latter target, I-IPW and CI-IPW achieve nearly identical outcomes (0.799 and 0.800), confirming the gains from personalization. These savings disappear at $T = 0.79$–0.80, as meeting higher targets requires treating most individuals, leaving little scope for selective allocation. I-HJ generally remains more treatment-intensive. Comparing method-specific and interference-aware outcome estimates helps explain these differences. NI-IPW slightly underestimates outcomes at $T = 0.77$–0.80, potentially contributing to its higher treatment intensity. In contrast, I-HJ's estimates are similar or slightly higher, so outcome underestimation cannot explain its greater resource use. The savings under I-IPW come with somewhat lower feasibility: at $T = 0.77$ and 0.78, its feasibility rate is 0.60 and 0.80, compared with 0.80–1.00 and 1.00 for the other proce-

Table 3: Out-of-sample comparison of I-IPW, I-HJ, NI-IPW, and CI-IPW policies.

Target	Policy	Avg. prob.	IPW outcome	Feasibility	Welfare loss	Own outcome
0.77	I-IPW	0.309	0.777	0.60	0.005	0.777
0.77	I-HJ	0.502	0.794	0.80	0.002	0.796
0.77	NI-IPW	0.401	0.791	0.80	0.006	0.790
0.77	CI-IPW	0.532	0.794	1.00	0.000	0.794
0.78	I-IPW	0.518	0.799	0.80	0.003	0.799
0.78	I-HJ	0.745	0.800	1.00	0.000	0.802
0.78	NI-IPW	0.620	0.806	1.00	0.000	0.803
0.78	CI-IPW	0.619	0.800	1.00	0.000	0.800
0.79	I-IPW	0.728	0.808	1.00	0.000	0.808
0.79	I-HJ	0.865	0.806	0.80	0.001	0.806
0.79	NI-IPW	0.718	0.807	0.80	0.001	0.802
0.79	CI-IPW	0.687	0.804	1.00	0.000	0.804
0.80	I-IPW	0.732	0.805	0.60	0.004	0.805
0.80	I-HJ	0.849	0.802	0.40	0.005	0.805
0.80	NI-IPW	0.721	0.802	0.60	0.007	0.799
0.80	CI-IPW	0.731	0.807	0.80	0.001	0.807
0.81	I-IPW	0.777	0.811	0.60	0.004	0.811
0.81	I-HJ	0.127	0.776	0.20	0.036	0.781
0.81	NI-IPW	0.272	0.778	0.20	0.034	0.780
0.81	CI-IPW	0.269	0.777	0.00	0.033	0.777

Notes: Averages over $K = 5$ held-out folds $\mathcal{I}_k$, with $\hat{p}^{-k}$ estimated on clusters outside $\mathcal{I}_k$. *Avg. prob.* is $K^{-1}\sum_k \widehat{\bar{C}}^k(\hat{p}^{-k})$, where $\widehat{\bar{C}}^k$ is the average treatment probability on $\mathcal{I}_k$; *IPW outcome* is $K^{-1}\sum_k \widehat{\bar{Y}}^k_{\text{I-IPW}}(\hat{p}^{-k})$, where $\widehat{\bar{Y}}^k_{\text{I-IPW}}$ is the interference-aware IPW outcome estimator on $\mathcal{I}_k$; *Feasibility* is $K^{-1}\sum_k \mathbf{1}\{\widehat{\bar{Y}}^k_{\text{I-IPW}}(\hat{p}^{-k}) \geq T\}$; *Welfare loss* is $K^{-1}\sum_k [T - \widehat{\bar{Y}}^k_{\text{I-IPW}}(\hat{p}^{-k})]_+$; *Own outcome* is $K^{-1}\sum_k \widehat{\bar{Y}}^k_m(\hat{p}^{-k})$, where $\widehat{\bar{Y}}^k_m$ estimates the outcome on $\mathcal{I}_k$ by the same estimator in the corresponding procedure, $m \in \{\text{I-IPW}, \text{I-HJ}, \text{NI-IPW}\}$.

dures. Nevertheless, the resulting welfare losses are 0.005 and 0.003, and remain below 0.007 across all procedures for targets up to $T = 0.80$. At $T = 0.81$, I-HJ, NI-IPW, and CI-IPW select low-treatment policies because the target generally cannot be attained in training, leading the optimization to favor minimal treatment. These policies are feasible in at most one fold, with welfare losses of 0.033–0.036, whereas I-IPW remains feasible in three folds, with a welfare loss of 0.004.

In-sample estimated policy allocations. Tables 4 and 5 report estimated treatment probabilities by covariate profile and averaged within covariate categories, respectively, for I-IPW, I-HJ and NI-IPW. For covariate X^k_{ij} at value x^k, the latter is the weighted average $\hat{p}_k(x^k) = \sum_{i,j} w_{ij}\,\hat{p}(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)/\sum_{i,j} w_{ij}$, with weights $w_{ij} = n_i^{-1}\mathbf{1}\{X^k_{ij} = x^k\}$, and analogously for cluster-level covariates. For brevity, we consider $T = (0.77, 0.78, 0.79)$. Table 4 shows substantial differences across covariate profiles. At $T = 0.77$, all three estimators assign the highest probabilities to high-SISBEN males in large households in Suba, and the lowest to low-SISBEN females in small households in San Cristóbal. At $T = 0.78$, I-IPW and NI-IPW

continue to prioritize males in large households in Suba, whereas I-HJ favors low-SISBEN males in small households in Suba. At $T = 0.79$, I-IPW and I-HJ prioritize low-SISBEN females in Suba, in large and small households respectively, while NI-IPW favors low-SISBEN students of either gender in large households in Suba.

Table 5 shows which groups receive more treatment on average. All estimators favor Suba across targets. I-IPW and NI-IPW consistently favor larger households, whereas I-HJ favors smaller households at $T = 0.78$–0.79. At $T = 0.77$, all three favor males and high-SISBEN students. At $T = 0.78$, males remain favored, but I-HJ and NI-IPW shift toward low-SISBEN students. By $T = 0.79$, both interference-aware estimators favor females and low-SISBEN students, whereas NI-IPW assigns nearly equal probabilities by gender. This shift may reflect that prioritizing males and high-SISBEN students is sufficient to meet lower targets, but becomes less effective at higher targets, when adverse spillovers on untreated females and lower attendance among low-SISBEN students may limit attainable outcomes. Meeting these higher targets may therefore require treating more females to mitigate negative spillovers and more low-SISBEN students to improve attendance.

Table 4: Treatment probabilities under I-IPW, I-HJ, and NI-IPW by profile.

				$T = 0.77$			$T = 0.78$			$T = 0.79$		
Location	SISBEN	Gender	HH size	I-IPW	I-HJ	NI-IPW	I-IPW	I-HJ	NI-IPW	I-IPW	I-HJ	NI-IPW
San Cristobal	Low	Female	Small	0.092	0.148	0.079	0.219	0.860	0.442	0.825	0.968	0.762
San Cristobal	Low	Female	Large	0.436	0.186	0.401	0.667	0.661	0.846	0.970	0.822	0.931
San Cristobal	Low	Male	Small	0.215	0.321	0.190	0.432	0.890	0.683	0.746	0.918	0.771
San Cristobal	Low	Male	Large	0.663	0.382	0.629	0.841	0.720	0.937	0.952	0.652	0.934
San Cristobal	High	Female	Small	0.128	0.321	0.190	0.241	0.692	0.247	0.635	0.918	0.541
San Cristobal	High	Female	Large	0.518	0.381	0.625	0.691	0.429	0.697	0.922	0.657	0.832
San Cristobal	High	Male	Small	0.285	0.562	0.389	0.463	0.749	0.471	0.519	0.805	0.554
San Cristobal	High	Male	Large	0.742	0.627	0.819	0.859	0.490	0.862	0.883	0.419	0.842
Suba	Low	Female	Small	0.647	0.759	0.610	0.836	0.969	0.935	0.982	0.982	0.977
Suba	Low	Female	Large	0.916	0.799	0.903	0.968	0.921	0.988	0.997	0.923	0.994
Suba	Low	Male	Small	0.833	0.896	0.809	0.932	0.976	0.975	0.972	0.954	0.978
Suba	Low	Male	Large	0.969	0.916	0.964	0.989	0.934	0.996	0.995	0.795	0.994
Suba	High	Female	Small	0.727	0.896	0.809	0.852	0.920	0.856	0.953	0.954	0.939
Suba	High	Female	Large	0.944	0.916	0.964	0.973	0.800	0.974	0.992	0.796	0.984
Suba	High	Male	Small	0.879	0.959	0.920	0.940	0.938	0.942	0.926	0.883	0.942
Suba	High	Male	Large	0.977	0.967	0.986	0.989	0.851	0.990	0.987	0.619	0.984

Table 5: Marginal implied treatment probabilities for I-IPW, I-HJ, and NI-IPW policies.

		$T = 0.77$			$T = 0.78$			$T = 0.79$		
Dimension	Group	I-IPW	I-HJ	NI-IPW	I-IPW	I-HJ	NI-IPW	I-IPW	I-HJ	NI-IPW
Gender	Female	0.297	0.336	0.324	0.441	0.744	0.542	0.824	0.894	0.761
Gender	Male	0.452	0.518	0.481	0.620	0.779	0.716	0.753	0.777	0.764
SISBEN	Low	0.326	0.306	0.302	0.503	0.831	0.695	0.855	0.887	0.835
SISBEN	High	0.421	0.544	0.501	0.555	0.691	0.561	0.723	0.786	0.690
Household size	Small	0.278	0.427	0.308	0.430	0.825	0.539	0.730	0.910	0.710
Household size	Large	0.611	0.421	0.634	0.778	0.601	0.850	0.937	0.652	0.894
Location	San Cristóbal	0.305	0.351	0.334	0.470	0.733	0.580	0.762	0.823	0.731
Location	Suba	0.796	0.889	0.823	0.899	0.934	0.924	0.959	0.918	0.958

Variance-adaptive margin Table 6 compares the uniform and variance-adaptive margins for I-IPW and I-HJ. Here column *Margin* is the average of the cross-validation selected margin across held-out folds. For I-IPW, the adaptive margin is smaller at $T = 0.77$–0.80 and slightly larger at $T = 0.81$; for I-HJ, it is smaller at every target. This pattern reflects the policy-specific variance scaling of the adaptive margin. For I-IPW, the adaptive margin selects a less resource-intensive policy at every target. Averaging the treatment share by threshold, it reduces the mean treatment share from 0.613 to 0.514. This reduction is accompanied by somewhat lower out-of-sample feasibility and a slightly higher mean welfare loss. The same trade-off is more pronounced for I-HJ. Overall, the variance-adaptive procedure successfully reduces the average treatment probability at the cost of somewhat lower feasibility. Appendix B reports the treatment probabilities by covariate category under variance-adaptive margins: as with uniform margins, I-IPW consistently assigns higher

Table 6: Out-of-sample comparison of uniform and variance-adaptive I-IPW and I-HJ.

Target	Policy	Avg. prob.	IPW outcome	Feasibility	Welfare loss	Own outcome	Margin
0.77	I-IPW-U	0.309	0.777	0.60	0.005	0.777	0.027
0.77	I-IPW-VA	0.257	0.776	0.80	0.005	0.776	0.022
0.77	I-HJ-U	0.502	0.794	0.80	0.002	0.796	0.030
0.77	I-HJ-VA	0.127	0.776	0.60	0.009	0.781	0.007
0.78	I-IPW-U	0.518	0.799	0.80	0.003	0.799	0.030
0.78	I-IPW-VA	0.365	0.782	0.60	0.006	0.782	0.020
0.78	I-HJ-U	0.745	0.800	1.00	0.000	0.802	0.026
0.78	I-HJ-VA	0.245	0.774	0.60	0.012	0.784	0.008
0.79	I-IPW-U	0.728	0.808	1.00	0.000	0.808	0.028
0.79	I-IPW-VA	0.497	0.793	0.60	0.004	0.793	0.017
0.79	I-HJ-U	0.865	0.806	0.80	0.001	0.806	0.021
0.79	I-HJ-VA	0.516	0.789	0.60	0.008	0.796	0.010
0.80	I-IPW-U	0.732	0.805	0.60	0.004	0.805	0.024
0.80	I-IPW-VA	0.689	0.805	0.60	0.004	0.805	0.016
0.80	I-HJ-U	0.849	0.802	0.40	0.005	0.805	0.015
0.80	I-HJ-VA	0.714	0.802	0.40	0.006	0.799	0.006
0.81	I-IPW-U	0.777	0.811	0.60	0.004	0.811	0.010
0.81	I-IPW-VA	0.763	0.810	0.60	0.005	0.810	0.011
0.81	I-HJ-U	0.127	0.776	0.20	0.036	0.781	0.017
0.81	I-HJ-VA	0.127	0.776	0.20	0.036	0.781	0.007

probabilities to larger households, whereas I-HJ favors smaller households at $T = 0.78$–0.80.

9 Conclusion

This paper develops a framework for learning resource-efficient individualized stochastic treatment policies under heterogeneous partial interference. By combining interference-aware estimation with a smooth approximation to the outcome constraint, we obtain policies that balance resource use and population feasibility, together with non-asymptotic regret guarantees for both IPW and self-normalized estimators. Our variance-adaptive margin further accounts for policy-specific estimation uncertainty and can substantially reduce resource use. Our simulation study and the Colombian cash-transfer application illustrate the practical importance of accounting for interference and treatment heterogeneity when designing policies to achieve prespecified population-level targets.

Our framework suggests three potential extensions. First, policy estimation could incorporate observed network information. This applies both under partial interference, with networks observed within clusters, and in a single interconnected network, where estimation and regret bounds must also account for network dependence (Leung, 2020; Viviano, 2024). Second, the framework could be extended to observational studies with unknown propensity scores, using flexible doubly robust estimators, requiring guarantees that account for nuisance estimation error (Lee et al., 2025; Park et al., 2024). Finally, building on Park et al. (2024), the framework could be extended to individualized policies subject to cluster-level outcome targets rather than a population-level target. This would require adapting the constraint surrogate to the cluster level, allowing personalized treatment allocation while enforcing cluster-specific outcome requirements.

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A Resource-Constrained Optimal Policies

A.1 The dual resource-constrained problem

The main outcome-target problem formulation minimizes the average treatment probability subject to a population outcome target. The complementary resource-constrained problem maximizes the average outcome over a policy class under a constraint on the average treatment probability

$$\max_{p\in\mathcal{P}} \bar{Y}(p) \quad \text{subject to} \quad \bar{C}(p) \leq B. \tag{A.1}$$

with $B \in [0,1]$. Consistent with the main text, define the population value functions

$$\bar{C}^{\star}(T) = \inf_{p\in\mathcal{P}:\,\bar{Y}(p)\geq T} \bar{C}(p), \qquad \bar{Y}^{\star}(B) = \sup_{p\in\mathcal{P}:\,\bar{C}(p)\leq B} \bar{Y}(p). \tag{A.2}$$

We next characterize the relationship between the outcome-target problem in Eq. 2 and the resource-constrained problem in Eq. A.1. Given a budget B, consider the outcome-target problem with target $T^{\star} = \bar{Y}^{\star}(B)$, the maximal outcome attainable under B. Every budget-constrained maximizer attains $T^{\star}$ with average treatment probability at most B, so $\bar{C}^{\star}(T^{\star}) \leq B$. Equality, and hence identical optimizer sets, holds if and only if every budget-constrained maximizer exhausts the budget: a maximizer p with $\bar{C}(p) < B$ attains $T^{\star}$ with a strictly lower average treatment probability, so the two problems have different solutions. The following condition rules this out.

Assumption A.1 (Local outcome growth)**.** *Fix a budget B for which the outcome-maximization problem has a solution. For every $p \in \mathcal{P}$ with $\bar{C}(p) < B$ and every $\varepsilon > 0$, there exists $p' \in \mathcal{P}$ such that*

$$\|p' - p\|_{\infty} < \varepsilon, \qquad \bar{Y}(p') > \bar{Y}(p). \tag{A.3}$$

Assumption A.1 is a local outcome growth condition on the attainable stochastic policies; it does not require welfare to be globally increasing in every treatment probability.

Theorem A.1 (Equivalence at matched thresholds)**.** *Suppose Assumption A.1 holds and the budget-constrained problem (A.1) admits an optimizer. Set $T^{\star} = \bar{Y}^{\star}(B)$. Then $\bar{C}^{\star}(T^{\star}) = B$, and*

$$\arg\max_{p\in\mathcal{P}:\,\bar{C}(p)\leq B} \bar{Y}(p) = \arg\min_{p\in\mathcal{P}:\,\bar{Y}(p)\geq T^{\star}} \bar{C}(p).$$

Assumption A.1 is sufficient, not necessary. For a fixed budget B with attained optima, $\bar{C}^{\star}\{\bar{Y}^{\star}(B)\} = B$ holds if and only if every maximizer under budget B exhausts that budget. One sufficient primitive condition for local outcome growth is that every nontrivial componentwise increase in a cluster treatment vector strictly improves its cluster-average

potential outcome, almost surely, *together with* a policy class that allows arbitrarily small, nontrivial componentwise upward perturbations on a set of clusters of positive probability whenever $\bar{C}(p) < B$. This condition concerns aggregate effects, so individual negative spillovers are permitted when outweighed by positive effects on other cluster members. By contrast, sufficiently adverse spillovers may create welfare-satiation regions.

For illustration, consider a cluster containing two siblings, with

$$Y_{i1}(a_1, a_2) = 0.60 + 0.20a_1 - 0.25a_2, \qquad Y_{i2}(a_1, a_2) = 0.60 + 0.15a_2.$$

For independent treatment probabilities $p = (p_1, p_2)$, the expected cluster-average outcome and intensity are

$$\bar{Y}(p) = 0.60 + 0.10p_1 - 0.05p_2, \qquad \bar{C}(p) = \frac{p_1 + p_2}{2}.$$

Restrict $p_1, p_2 \in [0.2, 0.8]$ and take $B = 0.75$. The budget-constrained maximizer is $p^\star = (0.8, 0.2)$, with $\bar{Y}^\star(0.75) = 0.67$ but $\bar{C}(p^\star) = 0.50$. Consequently, $\bar{C}^\star\{\bar{Y}^\star(0.75)\} = 0.50 < 0.75$. Thus the value-function inversion at $T = \bar{Y}^\star(B)$ and equality of optimizer sets can fail when a budget-constrained optimum leaves resources unused, even though the general relation $\bar{C}^\star(T) = \inf\{b : \bar{Y}^\star(b) \geq T\}$ continues to hold.

A.1.1 Estimation and regret for outcome maximization

Let $\widehat{\bar{Y}}^{\text{I-IPW}}(p)$ and $\widehat{\bar{C}}(p)$ be as in the main text. Define the reflected smooth hinge

$$h_\epsilon^+(c; B) = h_\epsilon(-c; -B).$$

It is zero for $c \leq B - \epsilon$, equals one for $c \geq B$, and is $2/\epsilon$-Lipschitz. Fix a penalty level $C_y > 2M$, where $|Y_{ij}(a_i)| \leq M$, and define

$$\begin{aligned} V_\epsilon^W(p) &= \bar{Y}(p) - C_y h_\epsilon^+\{\bar{C}(p); B\}, \\ \widehat{V}_\epsilon^W(p) &= \widehat{\bar{Y}}^{\text{I-IPW}}(p) - C_y h_\epsilon^+\{\widehat{\bar{C}}(p); B\}, \\ \widehat{p}_\epsilon^W &\in \arg\max_{p \in \mathcal{P}} \widehat{V}_\epsilon^W(p). \end{aligned}$$

The population penalized criterion and expected regret are

$$R^W(p) = \bar{Y}(p) - C_y \mathbf{1}\{\bar{C}(p) > B\}, \qquad \text{Reg}^W(\widehat{p}) = \bar{Y}^\star(B) - \mathbb{E}\{R^W(\widehat{p})\}.$$

Since $C_y > 2M$ and there is a budget-feasible policy, $\sup_{p \in \mathcal{P}} R^W(p) = \bar{Y}^\star(B)$.

Budget-side approximation error. An analogue of Theorem 2 controls the surrogate approximation error through a local direction of decreasing resource use. Let $p_B^\star$ attain $\bar{Y}^\star(B)$. If $\bar{C}(p_B^\star) < B$, then $\bar{Y}^\star(B - \epsilon) = \bar{Y}^\star(B)$ for all sufficiently small $\epsilon > 0$. Otherwise, suppose $\bar{C}(p_B^\star) = B$ and that there are constants $\kappa, L, t_0 > 0$ and a path $\{p_t : 0 \leq t \leq t_0\} \subseteq \mathcal{P}$

satisfying

$$p_0 = p_B^\star, \qquad \bar{C}(p_t) \le B - \kappa t, \qquad \|p_t - p_B^\star\|_\infty \le Lt \quad (0 \le t \le t_0). \tag{A.4}$$

Under Assumption 3, the population outcome functional is already Lipschitz with respect to the policy probabilities:

$$|\bar{Y}(p) - \bar{Y}(q)| \le 2Mn_{\max}\|p - q\|_\infty. \tag{A.5}$$

Indeed, couple each cluster's independent Bernoulli treatment vectors under p and q using the same independent uniforms. The probability that any treatment differs is at most $\sum_{j=1}^{n_i} |p(X_{ij}, V_i) - q(X_{ij}, V_i)|$, while a cluster-average potential outcome changes by at most $2M$ if the assignment vector changes. Taking expectations gives (A.5). For $0 < \epsilon \le \kappa t_0$, take $t = \epsilon/\kappa$. Then $\bar{C}(p_t) \le B - \epsilon$ and

$$0 \le \bar{Y}^\star(B) - \bar{Y}^\star(B - \epsilon) \le \bar{Y}(p_B^\star) - \bar{Y}(p_t) \le \frac{2Mn_{\max}L}{\kappa}\epsilon. \tag{A.6}$$

Hence the approximation error is $O(\epsilon)$ without assuming that $\mathcal{P}$ contains the zero-treatment policy or is closed under probability scaling.

Condition (A.4) is readily verified for a logistic policy family when the parameter space permits a small downward shift of the intercept. Writing $p_\gamma(x, v) = \sigma\{\gamma_0 + g_{\tilde{\gamma}}(x, v)\}$, set $p_t = p_{(\gamma_0^\star - t, \tilde{\gamma}^\star)}$. Since σ is 1/4-Lipschitz, $\|p_t - p_B^\star\|_\infty \le t/4$; furthermore,

$$\frac{\mathrm{d}}{\mathrm{d}t}\bar{C}(p_t)\bigg|_{t=0} = -\mathbb{E}_{G_0}\left\{\frac{1}{n_i}\sum_{j=1}^{n_i}\sigma'\big(\gamma_0^\star + g_{\tilde{\gamma}^\star}(X_{ij}, V_i)\big)\right\} < 0.$$

For finite predictors, continuity then provides a $\kappa > 0$ on a sufficiently short interval. Mere continuity of the policy function is not sufficient: the admissible class must allow the downward perturbation. Unlike the outcome-growth condition of Theorem 2, this property depends on the policy class rather than on the sign of direct or spillover treatment effects.

Theorem A.2 (Regret for resource-constrained policies)**.** *Suppose Assumptions 1–3 hold, $C_y > 2M$, and the relevant optimization problems admit solutions. Let $\mathfrak{R}_N = \mathfrak{R}_N(\mathcal{P})$.*

(i) General surrogate bound. *For $\epsilon > 0$ such that $\bar{Y}^\star(B - \epsilon)$ is defined,*

$$\mathrm{Reg}^W(\widehat{p}_\epsilon^W) \le \bar{Y}^\star(B) - \bar{Y}^\star(B - \epsilon) + 4\sqrt{2}\,\frac{M}{\eta}n_{\max}^{3/2}\mathfrak{R}_N + \frac{2M}{\eta}\sqrt{\frac{2\log 2}{N}} + \frac{16C_y n_{\max}}{\epsilon}\mathfrak{R}_N. \tag{A.7}$$

If the local downward-direction condition (A.4) holds and $\mathfrak{R}_N = O(N^{-1/2})$, then choosing $\epsilon \asymp N^{-1/4}$ gives $\mathrm{Reg}^W(\widehat{p}_\epsilon^W) = O(N^{-1/4})$.

(ii) Refined bound. *For* $\delta \in (0,1)$ *define*

$$b_C(\delta) = 2n_{\max}\mathfrak{R}_N + \sqrt{\frac{\log(4/\delta)}{2N}},$$
$$b_Y(\delta) = 2\sqrt{2}\frac{M}{\eta}n_{\max}^{3/2}\mathfrak{R}_N + \frac{M}{\eta}\sqrt{\frac{2\log(4/\delta)}{N}},$$
$$\alpha_N(\delta) = \frac{2M + 2b_Y(\delta)}{C_y},$$

and let

$$k(\alpha) = \begin{cases} \sqrt{(1-\alpha)/2}, & 1/2 \le \alpha < 1, \\ 1 - \sqrt{\alpha/2}, & 0 < \alpha < 1/2. \end{cases}$$

Suppose $\alpha_N(\delta) < 1$, *that* $\bar{Y}^\star\{B - \epsilon - b_C(\delta)\}$ *is defined, and that* $\epsilon\, k\{\alpha_N(\delta)\} \ge b_C(\delta)$. *Then*

$$\mathrm{Reg}^W(\widehat{p}_\epsilon^W) \le \bar{Y}^\star(B) - \bar{Y}^\star\{B - \epsilon - b_C(\delta)\} + 2b_Y(\delta) + (2M + C_y)\delta. \tag{A.8}$$

If the local downward-direction condition (A.4) *holds,* $\mathfrak{R}_N = O(N^{-1/2})$, *and* $C_y > 2M$ *is fixed, then* $\delta_N = N^{-1}$ *and a choice* $\epsilon_N \asymp \sqrt{\log N/N}$ *sufficiently large to satisfy the displayed margin condition yield* $\mathrm{Reg}^W(\widehat{p}_{\epsilon_N}^W) = O\{\sqrt{\log N/N}\}$.

The distinction between the two regret regimes parallels the main-text analysis. The general bound trades the left-hand frontier approximation error $\bar{Y}^\star(B) - \bar{Y}^\star(B-\epsilon)$ against an estimation term of order $\epsilon^{-1}\mathfrak{R}_N$. The refined bound instead uses uniform constraint concentration to certify the population budget feasibility of the learned policy. Importantly, it also incurs outcome-estimation error through the *objective*. Hence its favorable feasibility guarantee does not imply that learning the welfare-maximizing policy is free of interference-related statistical uncertainty. A Hájek analogue can be obtained under the strict-positivity conditions used in the main text by replacing the IPW deviation bound with the corresponding self-normalized bound.

A.2 Simulation study

We use the data-generating process and individualized logistic policy class of Section 7. The expected treatment-intensity budget is $B = 0.3185$, matching the oracle treatment share associated with the outcome target $T = 0.75$ in the main simulation. We examine interference-aware IPW and Hájek estimators with a uniform hinge margin, together with the no-interference IPW estimator (NI-IPW) and a constant-policy interference-aware estimator (CI-IPW). The margin $\epsilon_N = 0.1\sqrt{\log(N)/N}$ is held common across methods.

All four procedures satisfy the *population* expected-resource constraint in every one of the 500 Monte Carlo replications for each sample size considered. This is consistent with the simpler structure of the expected treatment-intensity constraint, which depends only

Table 7: Monte Carlo performance of the budget-constrained uniform-margin procedures and benchmarks.

N	Estimator	Regret	Cost	Outcome	Feasible	Margin
200	I-IPW	0.059 (0.051)	0.295 (0.027)	0.691 (0.051)	1.000 (0.000)	0.0163
	I-HJ	0.043 (0.030)	0.299 (0.015)	0.707 (0.030)	1.000 (0.000)	0.0163
	CI-IPW	0.086 (0.009)	0.301 (0.005)	0.664 (0.009)	1.000 (0.000)	0.0163
	NI-IPW	0.038 (0.006)	0.301 (0.003)	0.712 (0.006)	1.000 (0.000)	0.0163
400	I-IPW	0.042 (0.041)	0.301 (0.022)	0.708 (0.041)	1.000 (0.000)	0.0122
	I-HJ	0.032 (0.018)	0.304 (0.008)	0.718 (0.018)	1.000 (0.000)	0.0122
	CI-IPW	0.079 (0.004)	0.305 (0.003)	0.671 (0.004)	1.000 (0.000)	0.0122
	NI-IPW	0.031 (0.005)	0.305 (0.002)	0.719 (0.005)	1.000 (0.000)	0.0122
800	I-IPW	0.031 (0.027)	0.306 (0.014)	0.719 (0.027)	1.000 (0.000)	0.0091
	I-HJ	0.027 (0.013)	0.307 (0.006)	0.723 (0.013)	1.000 (0.000)	0.0091
	CI-IPW	0.074 (0.001)	0.308 (0.001)	0.676 (0.001)	1.000 (0.000)	0.0091
	NI-IPW	0.027 (0.005)	0.308 (0.002)	0.723 (0.005)	1.000 (0.000)	0.0091
2000	I-IPW	0.023 (0.007)	0.311 (0.002)	0.727 (0.007)	1.000 (0.000)	0.0062
	I-HJ	0.024 (0.006)	0.310 (0.002)	0.726 (0.006)	1.000 (0.000)	0.0062
	CI-IPW	0.069 (0.002)	0.311 (0.001)	0.681 (0.002)	1.000 (0.000)	0.0062
	NI-IPW	0.022 (0.004)	0.310 (0.002)	0.728 (0.004)	1.000 (0.000)	0.0062

Notes: Monte Carlo means (standard deviations) over 500 replications. Cost is $\bar{C}(\hat{p})$; Outcome is $\bar{Y}(\hat{p})$; Feasible is $1\{\bar{C}(\hat{p}) \leq B\}$. Regret is $R^{W\star}_{G_0} - R^{W}_{G_0}(\hat{p})$, where $R^{W}(p) = \bar{Y}(p) - C_y 1\{\bar{C}(p) > B\}$, $C_y = 50$, $R^{W\star}_{G_0} = 0.7500$, and $B = 0.3185$. The uniform margin is $0.1\sqrt{\log(N)/N}$. The reference is the population oracle at budget B, without subtracting the finite-sample margin.

on observed covariates and policy probabilities. The feasibility finding must, however, be interpreted alongside welfare: at $N = 2000$ the average outcomes under interference-aware IPW and Hájek are 0.727 and 0.726, respectively, below the oracle value $\bar{Y}^{\star}(B) = 0.750$. The corresponding average treatment intensities, 0.311 and 0.310, are also below the budget $B = 0.3185$. Part of the welfare loss therefore reflects conservative use of the available expected budget. As the sample size grows, treatment intensity approaches B and welfare regret generally declines.

A second difference from the outcome-target formulation concerns misspecification of interference. In the present budget-constrained simulation, NI-IPW attains welfare close to the interference-aware procedures: at $N = 2000$, its mean outcome is 0.728, versus 0.727 and 0.726 for interference-aware IPW and Hájek. This result suggests that, in this design and at this budget, ignoring interference does not substantially change the population welfare attained by the selected policy. The distinction is more pronounced in the outcome-target simulation of Section 7, where NI-IPW incorrectly classifies policies as meeting the outcome target and its population feasibility rate falls to 0.142 at $N = 2000$. In budget-constrained welfare maximization, ignoring interference biases the estimated *objective* but does not bias the expected-intensity constraint; good policy selection remains possible if the misspecified objective preserves enough information about the relative welfare of competing policies. In

outcome-constrained resource minimization, the same misspecification directly distorts the *estimated feasible set*, potentially causing systematic target violations.

Finally, CI-IPW achieves an average outcome of 0.681 at $N = 2000$, compared with 0.727 for individualized interference-aware IPW, illustrating the welfare gains from covariate-based personalization under the same expected-resource budget.

A.3 Colombian CCT application under a budget constraint

Table 8 reports held-out performance of the budget-constrained estimators for budgets $B \in \{0.35, 0.50, 0.65, 0.80\}$; All methods satisfy the held-out expected-resource constraint in every reported outer fold, with no observed budget excess. Higher budgets are associated with greater treatment intensity and generally higher estimated outcomes: for interference-aware IPW, the treatment share rises from 0.328 to 0.768, while its held-out IPW-evaluated outcome rises from 0.783 to 0.812. At a given budget, the estimated outcomes of the individualized interference-aware and no-interference policies are fairly similar under the common IPW evaluation column, consistent with the simulation findings. These are estimated, not known population, outcomes; the comparison therefore does not show that interference can be ignored generally.

Budget	Policy	Treat. Prob	Own outcome	I-IPW outcome	Budget feasibility
0.35	I-IPW	0.328	0.783	0.783	1.00
0.35	I-HJ	0.328	0.793	0.785	1.00
0.35	NI-IPW	0.328	0.790	0.790	1.00
0.35	CI-IPW	0.329	0.780	0.780	1.00
0.50	I-IPW	0.469	0.799	0.799	1.00
0.50	I-HJ	0.477	0.797	0.788	1.00
0.50	NI-IPW	0.462	0.793	0.797	1.00
0.50	CI-IPW	0.479	0.790	0.790	1.00
0.65	I-IPW	0.587	0.802	0.802	1.00
0.65	I-HJ	0.627	0.798	0.796	1.00
0.65	NI-IPW	0.580	0.796	0.800	1.00
0.65	CI-IPW	0.629	0.800	0.800	1.00
0.80	I-IPW	0.768	0.812	0.812	1.00
0.80	I-HJ	0.777	0.803	0.807	1.00
0.80	NI-IPW	0.774	0.807	0.811	1.00
0.80	CI-IPW	0.779	0.810	0.810	1.00

Table 8: Out-of-sample Colombia budget-constraint results under uniform margins. Feasibility is the fraction of outer folds satisfying the budget.

B Additional Tables

Table 9: Marginal implied treatment probabilities under I-IPW-VA and I-HJ-VA.

Dimension	Group	Policy type	T=0.77	T=0.78	T=0.79	T=0.80	T=0.81
Gender	Female	I-IPW-VA	0.244	0.346	0.368	0.767	0.836
Gender	Female	I-HJ-VA	0.099	0.186	0.390	0.522	0.099
Gender	Male	I-IPW-VA	0.399	0.499	0.547	0.772	0.761
Gender	Male	I-HJ-VA	0.216	0.329	0.592	0.720	0.216
SISBEN	Low SISBEN	I-IPW-VA	0.329	0.402	0.430	0.843	0.864
SISBEN	Low SISBEN	I-HJ-VA	0.104	0.164	0.367	0.641	0.104
SISBEN	High SISBEN	I-IPW-VA	0.312	0.440	0.483	0.695	0.734
SISBEN	High SISBEN	I-HJ-VA	0.209	0.349	0.611	0.598	0.209
HH size	Small	I-IPW-VA	0.196	0.304	0.352	0.711	0.744
HH size	Small	I-HJ-VA	0.095	0.332	0.508	0.653	0.095
HH size	Large	I-IPW-VA	0.632	0.716	0.718	0.915	0.937
HH size	Large	I-HJ-VA	0.311	0.068	0.443	0.536	0.311
Location	San Cristobal	I-IPW-VA	0.368	0.402	0.402	0.746	0.775
Location	San Cristobal	I-HJ-VA	0.163	0.207	0.430	0.566	0.163
Location	Suba	I-IPW-VA	0.027	0.542	0.796	0.910	0.947
Location	Suba	I-HJ-VA	0.117	0.564	0.856	0.953	0.117

C Proofs

This section provides the derivations of Theorem 1-5 in the main text, as well as Theorem C.3 for the regret bound on the Hajek policy estimator with variance-adaptive margin, and the results in Appendix A. We start by introducing some preliminary lemmas, and then move to the main proofs.

C.1 Preliminary Lemmas and Propositions

Lemma C.1 (Uniform deviation bounds for the average treatment probability)**.** *Suppose Assumption 3.b) holds. For any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\sup_{p\in\mathcal{P}} |\bar{C}(p) - \widehat{\bar{C}}(p)| \leq 2n_{\max}\mathcal{R}_N(\mathcal{P}) + \sqrt{\frac{\log(2/\delta)}{2N}}.$$

Moreover,

$$\mathbb{E}_{\mathcal{O}}\left[\sup_{p\in\mathcal{P}} |\bar{C}(p) - \widehat{\bar{C}}(p)|\right] \leq 4n_{\max}\mathcal{R}_N(\mathcal{P}).$$

Proof. Let $f_p(\boldsymbol{O}_i) = n_i^{-1}\sum_{j=1}^{n_i} p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)$ and $\mathcal{F}' = \{f_p : p \in \mathcal{P}\}$. Then $0 \leq f_p \leq 1$, $\widehat{\bar{C}}(p) = N^{-1}\sum_i f_p(\boldsymbol{O}_i)$, and $\bar{C}(p) = \mathbb{E}_{G_0} f_p(\boldsymbol{O}_i)$. The one-sided symmetrization inequality (Shalev-Shwartz and Ben-David, 2014, Lemma 26.2) gives

$$\mathbb{E}_{\mathcal{O}} \sup_{p\in\mathcal{P}}\{\bar{C}(p) - \widehat{\bar{C}}(p)\} \leq 2\mathcal{R}_N(\mathcal{F}'). \tag{C.1}$$

We bound $R_N(\mathcal{F}')$ as follows. Write

$$R_N(\mathcal{F}') = \mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N}\sigma_i \frac{1}{n_i}\sum_{j=1}^{n_i} p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)\right].$$

and pad each cluster to length $n_{\max}$, with terms for $j > n_i$ set to zero. Then

$$\frac{1}{n_i}\sum_{j=1}^{n_i} p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i) = \sum_{j=1}^{n_{\max}} \frac{\mathbf{1}\{j \le n_i\}}{n_i}\, p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i),$$

so

$$R_N(\mathcal{F}') = \mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N}\sigma_i \sum_{j=1}^{n_{\max}} \frac{\mathbf{1}\{j \le n_i\}}{n_i}\, p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)\right].$$

Applying the triangle inequality,

$$\begin{aligned} R_N(\mathcal{F}') &\le \sum_{j=1}^{n_{\max}} \mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N}\sigma_i \frac{\mathbf{1}\{j \le n_i\}}{n_i}\, p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)\right] \\ &\le \sum_{j=1}^{n_{\max}} \mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N}\sigma_i p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)\right] \\ &= n_{max} R_N(\mathcal{P}). \end{aligned}$$

since $0 \le \mathbf{1}\{j \le n_i\}/n_i \le 1$. Combining this calculation with (C.1) yields $\mathbb{E}_{\mathcal{O}} \sup_p\{\bar{C}(p) - \widehat{\bar{C}}(p)\} \le 2n_{\max}\mathcal{R}_N(\mathcal{P})$. By Rademacher symmetry, the same bound holds in the opposite direction. Write $Z_C^+ = \sup_p\{\bar{C}(p) - \widehat{\bar{C}}(p)\}$ and $Z_C^- = \sup_p\{\widehat{\bar{C}}(p) - \bar{C}(p)\}$. Changing one cluster changes either supremum by at most $1/N$. Thus McDiarmid's inequality and a union bound give the stated high-probability result. Since $\mathcal{P}$ contains a constant policy, both Z_C^+ and Z_C^- are nonnegative, and $\mathbb{E}_{\mathcal{O}} \max\{Z_C^+, Z_C^-\} \le \mathbb{E}_{\mathcal{O}}(Z_C^+ + Z_C^-) \le 4n_{\max}\mathcal{R}_N(\mathcal{P})$, proving the expected bound. □

Lemma C.2 (Uniform deviation bounds for the average outcome)**.** *Suppose Assumptions 1–3 hold. For any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\sup_{p\in\mathcal{P}} |\widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p)| \le 2\sqrt{2}\frac{M}{\eta} n_{\max}^{3/2}\mathcal{R}_N(\mathcal{P}) + \frac{M}{\eta}\sqrt{\frac{2\log(2/\delta)}{N}}$$

Moreover,

$$\mathbb{E}_{\mathcal{O}}\left[\sup_{p\in\mathcal{P}} |\widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p)|\right] \le 2\sqrt{2}\frac{M}{\eta} n_{\max}^{3/2}\mathcal{R}_N(\mathcal{P}) + \frac{M}{\eta}\sqrt{\frac{2\log 2}{N}}.$$

Proof. Let $\psi^{\text{I-IPW}}(\boldsymbol{O}_i, p) = n_i^{-1}\sum_{j=1}^{n_i} Y_{ij}(\boldsymbol{A}_i)\pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p)/f(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i)$ and $\mathcal{F}'' = \{\psi^{\text{I-IPW}}(\cdot, p) : p \in \mathcal{P}\}$. By identification, $\bar{Y}(p) = \mathbb{E}_{G_0}\psi^{\text{I-IPW}}(\boldsymbol{O}_i, p)$ and $\widehat{\bar{Y}}_{\text{I-IPW}}(p) = N^{-1}\sum_i \psi^{\text{I-IPW}}(\boldsymbol{O}_i, p)$.

One-sided symmetrization gives

$$\mathbb{E}_{\mathcal{O}} \sup_{p\in\mathcal{P}}\{\bar{Y}(p) - \widehat{\bar{Y}}_{\text{I-IPW}}(p)\} \le 2\mathcal{R}_N(\mathcal{F}''). \tag{C.2}$$

We now bound $R_N(\mathcal{F}'')$. Write

$$\psi^{\text{I-IPW}}(\boldsymbol{O}_i, p) = W_i\, \pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p), \quad W_i = \frac{1}{n_i}\sum_{j=1}^{n_i} \frac{Y_{ij}(\boldsymbol{A}_i)}{f(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i)}.$$

By bounded outcomes and overlap, $|W_i| \le \frac{M}{\eta}$. Hence, by contraction,

$$R_N(\mathcal{F}'') \le \frac{M}{\eta}\,\mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N} \sigma_i \pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p)\right].$$

Let

$$\pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p) = \prod_{j=1}^{n_i} q_{ij}(p), \quad q_{ij}(p) = p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i)^{A_{ij}}(1 - p(\boldsymbol{X}_{ij}, \boldsymbol{V}_i))^{1-A_{ij}}.$$

The map $u \mapsto \prod_{j=1}^{n_i} u_j$ is $\sqrt{n_i}$-Lipschitz w.r.t. ℓ_2, so by Corollary 4 in Maurer (2016),

$$\begin{aligned} R_N(\mathcal{F}'') &\le \frac{M}{\eta}\sqrt{2n_{\max}}\,\mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{n_i} \sigma_{ij} q_{ij}(p)\right] \\ &\le \frac{M}{\eta}\sqrt{2n_{\max}}\sum_{j=1}^{n_{\max}} \mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N} \sigma_{ij}\mathbf{1}\{j \le n_i\} q_{ij}(p)\right]. \end{aligned}$$

For each $j = 1, \ldots, n_i$, $q_{ij}(p)$, is $1-$Lipschitz in p. Hence, by contraction:

$$\mathbb{E}_{\mathcal{O},\sigma}\left[\sup_{p\in\mathcal{P}} \frac{1}{N}\sum_{i=1}^{N} \sigma_{ij}\mathbf{1}\{j \le n_i\} q_{ij}(p)\right] \le R_N(\mathcal{P}).$$

Therefore,

$$R_N(\mathcal{F}'') \le \frac{M}{\eta}\sqrt{2n_{\max}} \cdot n_{\max} R_N(\mathcal{P}) = \frac{M}{\eta}\sqrt{2}\, n_{\max}^{3/2} R_N(\mathcal{P}).$$

Combining with (C.2) and using Rademacher symmetry in the opposite direction, the one-sided deviations $Z_Y^{\pm} := \sup_{p\in\mathcal{P}}\{\pm(\widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p))\}$ satisfy $\mathbb{E}_{\mathcal{O}} Z_Y^{\pm} \le A_{\text{I-IPW},N}$. Since $|\psi^{\text{I-IPW}}(\boldsymbol{O}_i, p)| \le M/\eta$, replacing a cluster changes either $Z_Y^{\pm}$ by at most $2M/(N\eta)$. McDiarmid's inequality and a union bound give the stated high-probability inequality. Moreover, its exponential-moment form and the log-sum-exp inequality give

$$\begin{aligned} \mathbb{E}_{\mathcal{O}} \max\{Z_Y^+, Z_Y^-\} &\le 2\sqrt{2}\frac{M}{\eta} n_{\max}^{3/2}\mathcal{R}_N(\mathcal{P}) + \inf_{\lambda>0}\left\{\frac{\log 2}{\lambda} + \frac{\lambda M^2}{2N\eta^2}\right\} \\ &= 2\sqrt{2}\frac{M}{\eta} n_{\max}^{3/2}\mathcal{R}_N(\mathcal{P}) + \frac{M}{\eta}\sqrt{\frac{2\log 2}{N}}, \end{aligned}$$

which proves the expected bound. □

Lemma C.3 (Uniform deviation bounds for the Hájek average outcome)**.** *Suppose Assumptions 1–3 hold and $0 < p(x,v) < 1$ for all $p \in \mathcal{P}$ and $(x,v) \in \mathcal{X} \times \mathcal{V}$. For any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\sup_{p\in\mathcal{P}} |\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)| \le A_{\text{I-HJ},N} + \frac{M}{\eta}\sqrt{\frac{2\log(2/\delta)}{N}}, \qquad A_{\text{I-HJ},N} := 6\sqrt{2}\frac{M}{\eta} n_{\max}^{3/2}\mathcal{R}_N(\mathcal{P}) + \frac{2M}{\eta\sqrt{N}}.$$

Moreover,

$$\mathbb{E}_{\mathcal{O}}\left[\sup_{p\in\mathcal{P}} |\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)|\right] \le A_{\text{I-HJ},N} + \frac{M}{\eta}\sqrt{\frac{2\log 2}{N}}.$$

Proof. Recall $w_i(p) = \pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p)/f(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i)$ and $D_N(p) = N^{-1}\sum_i w_i(p)$. The strict positivity condition implies $w_i(p) > 0$ and hence $D_N(p) > 0$ for every $p \in \mathcal{P}$ and every N. Let us write

$$Ph = \mathbb{E}_{G_0} h(\boldsymbol{O}_i), \quad P_N h = N^{-1}\sum_i h(\boldsymbol{O}_i).$$

For $m \in [-M, M]$, let $g_{p,m}(\boldsymbol{O}_i) = w_i(p)(\bar{Y}_i - m)$ and $\mathcal{G}_H = \{g_{p,m} : p \in \mathcal{P},\ m \in [-M,M]\}$. Since $Pw_p = 1$ and $P(w_p\bar{Y}_i) = \bar{Y}(p)$, we have $Pg_{p,m} = \bar{Y}(p) - m$. Also, $\widehat{\bar{Y}}_{\text{I-HJ}}(p) \in [-M, M]$ and $P_N g_{p,\widehat{\bar{Y}}_{\text{I-HJ}}(p)} = 0$, so

$$\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p) = (P_N - P)g_{p,\widehat{\bar{Y}}_{\text{I-HJ}}(p)}. \tag{C.3}$$

Let $Z_H^+ = \sup_{g\in\mathcal{G}_H}(P_N - P)g$ and $Z_H^- = \sup_{g\in\mathcal{G}_H}(P - P_N)g$ denote the two one-sided uniform deviations, and let $Z_H = \max\{Z_H^+, Z_H^-\}$. We then have that, $\sup_p |\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)| \le \max\{Z_H^+, Z_H^-\}$. By one-sided symmetrization and Rademacher symmetry, $\mathbb{E}_{\mathcal{O}} Z_H^{\pm} \le 2\mathcal{R}_N(\mathcal{G}_H)$.

Let $\mathcal{F}_Y = \{\boldsymbol{O}_i \mapsto w_i(p)\bar{Y}_i : p \in \mathcal{P}\}$ and $\mathcal{W} = \{\boldsymbol{O}_i \mapsto w_i(p) : p \in \mathcal{P}\}$. Writing $S_w(p) = N^{-1}\sum_i \sigma_i w_i(p)$, the supremum over $m \in [-M, M]$ gives $\mathcal{R}_N(\mathcal{G}_H) \le \mathcal{R}_N(\mathcal{F}_Y) + M\mathbb{E}_{\mathcal{O},\sigma}\sup_p |S_w(p)|$. For any fixed $p_0 \in \mathcal{P}$, Rademacher symmetry and $0 \le w_i(p) \le 1/\eta$ imply

$$\mathbb{E}_{\mathcal{O},\sigma}\sup_p |S_w(p)| \le 2\mathcal{R}_N(\mathcal{W}) + \mathbb{E}_{\mathcal{O},\sigma}|S_w(p_0)| \le 2\mathcal{R}_N(\mathcal{W}) + \frac{1}{\eta\sqrt{N}}.$$

The vector-contraction bound used in Lemma C.2 gives $\mathcal{R}_N(\mathcal{F}_Y) \le \sqrt{2}(M/\eta) n_{\max}^{3/2}\mathcal{R}_N(\mathcal{P})$ and $\mathcal{R}_N(\mathcal{W}) \le \sqrt{2}\eta^{-1} n_{\max}^{3/2}\mathcal{R}_N(\mathcal{P})$. Therefore $\mathbb{E}_{\mathcal{O}} Z_H^{\pm} \le A_{\text{I-HJ},N}$.

For each p, m, the range of $g_{p,m}$ has width at most $2M/\eta$. Thus changing one cluster alters either $Z_H^{\pm}$ by at most $2M/(N\eta)$. McDiarmid's inequality and a union bound give the high-probability statement. Finally, the exponential-moment bounded-differences inequality and the log-sum-exp bound imply

$$\mathbb{E}_{\mathcal{O}}\max\{Z_H^+, Z_H^-\} \le A_{\text{I-HJ},N} + \inf_{\lambda>0}\left\{\frac{\log 2}{\lambda} + \frac{\lambda M^2}{2N\eta^2}\right\} = A_{\text{I-HJ},N} + \frac{M}{\eta}\sqrt{\frac{2\log 2}{N}},$$

which proves the expected bound. □

Lemma C.4 (Margin)**.** *Let $C_0 > 1$, let $\widetilde{Y} : \mathcal{P} \to \mathbb{R}$ be any estimator of $\bar{Y}$, let $m : \mathcal{P} \to (0, \infty)$ be any, possibly data-dependent, margin, and let*

$$\hat{p} \in \arg\min_{p \in \mathcal{P}} \left\{ \widehat{\bar{C}}(p) + C_0 h_{m(p)}(\widetilde{Y}(p); T) \right\}.$$

If $q \in \mathcal{P}$ satisfies $\widetilde{Y}(q) \geq T + m(q)$, then (a) $\widehat{\bar{C}}(\hat{p}) \leq \widehat{\bar{C}}(q)$; (b) $\widetilde{Y}(\hat{p}) \geq T + c\, m(\hat{p})$, where $c = c(C_0)$ is defined in Section 4.2.

Proof. Since $\widetilde{Y}(q) \geq T + m(q)$, we have $h_{m(q)}(\widetilde{Y}(q); T) = 0$. By optimality of $\hat{p}$,

$$\widehat{\bar{C}}(\hat{p}) + C_0 h_{m(\hat{p})}(\widetilde{Y}(\hat{p}); T) \leq \widehat{\bar{C}}(q).$$

Since the penalty is nonnegative, (a) follows. Moreover, since $0 \leq \widehat{\bar{C}}(p) \leq 1$,

$$h_{m(\hat{p})}(\widetilde{Y}(\hat{p}); T) \leq \frac{1}{C_0} < 1.$$

We now consider two cases. If $1 < C_0 \leq 2$, then $1/C_0 \geq 1/2$. By the definition of h_ϵ, if $\widetilde{Y}(\hat{p}) < T + m(\hat{p})/2$, then

$$1 - \frac{2}{m(\hat{p})^2} \big(\widetilde{Y}(\hat{p}) - T\big)^2 \leq \frac{1}{C_0},$$

which implies

$$\widetilde{Y}(\hat{p}) \geq T + m(\hat{p}) \sqrt{\frac{1 - 1/C_0}{2}}.$$

The same inequality holds if $\widetilde{Y}(\hat{p}) \geq T + m(\hat{p})/2$, since $\sqrt{(1 - 1/C_0)/2} \leq 1/2$.

If $C_0 \geq 2$, then $1/C_0 \leq 1/2$, which implies $\widetilde{Y}(\hat{p}) \geq T + m(\hat{p})/2$. If $\widetilde{Y}(\hat{p}) < T + m(\hat{p})$, the definition of h_ϵ gives

$$\frac{2}{m(\hat{p})^2} \big(T + m(\hat{p}) - \widetilde{Y}(\hat{p})\big)^2 \leq \frac{1}{C_0},$$

and therefore

$$\widetilde{Y}(\hat{p}) \geq T + m(\hat{p}) \left(1 - \frac{1}{\sqrt{2C_0}}\right).$$

This inequality also holds when $\widetilde{Y}(\hat{p}) \geq T + m(\hat{p})$. Thus, in both cases, $\widetilde{Y}(\hat{p}) \geq T + c\, m(\hat{p})$, establishing (b). □

Bernstein preliminaries. For a class $\mathcal{F}$ and $r > 0$, write $N_\infty(r, \mathcal{F}, k) = \sup_{\boldsymbol{o} \in \mathcal{O}^k} N(r, \mathcal{F}(\boldsymbol{o}), \|\cdot\|_\infty)$, where the covering centers belong to $\mathcal{F}(\boldsymbol{o})$. Let $Pf = \mathbb{E}_{G_0} f(O)$, $P_N f = N^{-1} \sum_i f(O_i)$ and $s_N(f)^2 = (N-1)^{-1} \sum_i \{f(O_i) - P_N f\}^2$. All the Bernstein results below require $N \geq 16$.

Theorem C.1 (Uniform empirical Bernstein bounds)**.** *Suppose Assumptions 1–3 hold and $N \geq 16$. Set $B_\psi = M/\eta$, $\Psi = \{\psi^{\text{I-IPW}}(\cdot, p) : p \in \mathcal{P}\}$ and $\widetilde{\Psi} = \{(\psi + B_\psi)/(2B_\psi) : \psi \in \Psi\}$.*

Define the common complexity term

$$\Lambda_{N,\text{I-IPW}}(\delta) = \log\{20N_\infty(1/N, \widetilde{\Psi}, 2N)/\delta\}.$$

With probability at least $1-\delta$*, simultaneously for all* $p \in \mathcal{P}$*,*

$$|\widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p)| \le \sqrt{18\, s_N(\psi_p^{\text{I-IPW}})^2 \Lambda_{N,\text{I-IPW}}(\delta)/N} + 32 B_\psi \Lambda_{N,\text{I-IPW}}(\delta)/N.$$

Moreover,

$$\Lambda_{N,\text{I-IPW}}(\delta) \le \log\{20N_\infty(2/(Nn_{\max}), \mathcal{P}, 2Nn_{\max})/\delta\}. \tag{C.4}$$

Proof. Bounded outcomes and overlap give $|\psi_p^{\text{I-IPW}}| \le B_\psi$, so $\widetilde{\psi}_p = (\psi_p^{\text{I-IPW}} + B_\psi)/(2B_\psi) \in [0,1]$. Apply Theorem 6 of Maurer and Pontil (2009) separately to $\widetilde{\Psi}$ and $1-\widetilde{\Psi}$, each at level $\delta/2$. Their covering numbers and empirical variances agree. A union bound therefore gives, uniformly in p,

$$|(P - P_N)\widetilde{\psi}_p| \le \sqrt{18\, s_N(\widetilde{\psi}_p)^2 \Lambda_{N,\text{I-IPW}}(\delta)/N} + 15\Lambda_{N,\text{I-IPW}}(\delta)/(N-1).$$

Multiplying by $2B_\psi$ and using $30/(N-1) \le 32/N$ for $N \ge 16$ proves the deviation bound.

For the covering bound, fix $2N$ clusters and an r-cover of $\mathcal{P}$ on their at most $2Nn_{\max}$ covariate pairs. The telescoping product inequality gives $|\pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; p) - \pi_i(\boldsymbol{A}_i \mid \boldsymbol{X}_i, \boldsymbol{V}_i; q)| \le n_{\max} r$ and hence $|\widetilde{\psi}_p(O_i) - \widetilde{\psi}_q(O_i)| \le n_{\max} r/2$. Taking $r = 2/(Nn_{\max})$ proves the claim. □

Lemma C.5 (Uniform bounds of empirical standard deviations)**.** *Under the assumptions of Theorem C.1, let* $\sigma_{\text{I-IPW}}(p)^2 = \text{Var}_{G_0}\{\psi_p^{\text{I-IPW}}(O)\}$ *and* $\widehat{\sigma}_{\text{I-IPW}}(p) = s_N(\psi_p^{\text{I-IPW}})$*. With probability at least* $1-\delta$*,*

$$\sup_{p\in\mathcal{P}} |\widehat{\sigma}_{\text{I-IPW}}(p) - \sigma_{\text{I-IPW}}(p)| \le 16 B_\psi \sqrt{\Lambda_{N,\text{I-IPW}}(\delta)/N}.$$

Proof. We give the uniform argument for any class $\mathcal{F} \subset [0,1]$. Set $K = N_\infty(1/N, \mathcal{F}, 2N)$ and $\kappa_N = \sqrt{N/(N-1)}$. For vectors u, v, centering is a contraction in Euclidean norm, so $|s_N(u) - s_N(v)| \le \kappa_N \|u - v\|_\infty$.

Let O' be an independent sample and $\sigma(f) = \sqrt{\text{Var}\{f(O)\}}$. By both tails of Theorem 10 of Maurer and Pontil (2009), for any fixed f the event $|s_N(f;O') - \sigma(f)| \le t/2$ has probability at least $1/2$ whenever $t/2 \ge \sqrt{2\log 4/(N-1)}$. Choosing a function witnessing a deviation (or an arbitrarily close witness), and conditioning on O, gives

$$\Pr\{\sup_f |s_N(f;O) - \sigma(f)| > t\} \le 2\Pr\{\sup_f |s_N(f;O) - s_N(f;O')| > t/2\}.$$

Condition on the paired observations and swap each pair independently with probability $1/2$. For each function, the two resulting vectors have independent coordinates and identical marginal distributions; their expected sample variances are equal. Theorem 10 and a

union bound over its two tails for both vectors give $\Pr_{\text{swap}}\{|s_N(f;O) - s_N(f;O')| > u\} \le 4\exp\{-(N-1)u^2/8\}$. A $1/N$-cover on the combined sample has at most K centers, and approximation changes the difference of standard deviations by at most $2\kappa_N/N$. Thus, for $t/2 > 2\kappa_N/N$,

$$\Pr\{\sup_f |s_N(f;O) - \sigma(f)| > t\} \le 8K\exp\{-(N-1)(t/2 - 2\kappa_N/N)^2/8\}.$$

Take $t = 4\kappa_N/N + \sqrt{32\log(8K/\delta)/(N-1)}$. This satisfies the preceding conditions and gives failure probability at most δ. For $N \ge 16$, $t \le 8\sqrt{\log(20K/\delta)/N}$. Apply this result to $\widetilde{\Psi}$ and multiply by $2B_\psi$. As usual, the suprema are understood to be measurable. □

Hájek Bernstein preliminaries. Assume throughout these Hájek results that $0 < p(\boldsymbol{x}, \boldsymbol{v}) < 1$ for every $p \in \mathcal{P}$ and covariate pair (x, v). As in Lemma C.3, this implies $w_i(p) > 0$ and $D_N(p) > 0$ for every sample. Reuse its class $\mathcal{G}_H = \{g_{p,m} : p \in \mathcal{P},\ m \in [-M, M]\}$, where $g_{p,m}(O_i) = w_i(p)(\bar{Y}_i - m)$, and its identity (C.3).

Define $\phi_i^{\text{I-HJ}}(p) = g_{p,\bar{Y}(p)}(O_i)$ and $\sigma_{\text{I-HJ}}(p)^2 = \text{Var}_{G_0}\{\phi_i^{\text{I-HJ}}(p)\}$. Let $\widetilde{\phi_i^{\text{I-HJ}}}(p) = g_{p,\widehat{\bar{Y}}_{\text{I-HJ}}(p)}(O_i)$. Its empirical mean is zero, and

$$\widetilde{\sigma}_{H,N}(p)^2 = s_N(\widetilde{\phi}^{\text{I-HJ}}(p))^2 = \frac{1}{N-1}\sum_i w_i(p)^2\{\bar{Y}_i - \widehat{\bar{Y}}_{\text{I-HJ}}(p)\}^2. \tag{C.5}$$

Thus $\widetilde{\sigma}_{H,N}(p) = D_N(p)\widehat{\sigma}_{\text{I-HJ}}(p)$. Also set $\widehat{\sigma}_w(p) = s_N(w_p)$ and $B_{\phi,H} = 2M/\eta$. Each $g_{p,m}$ has range contained in $[-(M+m)/\eta, (M-m)/\eta]$, of width $B_{\phi,H}$. Its normalized version is $\widetilde{g}_{p,m} = g_{p,m}/B_{\phi,H} + (M+m)/(2M) \in [0,1]$; write $\widetilde{\Phi}_H^\star = \{\widetilde{g}_{p,m} : p \in \mathcal{P},\ m \in [-M, M]\}$.

Theorem C.2 (Uniform empirical Bernstein bound for Hájek)**.** *Suppose Assumptions 1–3 hold, $N \ge 16$, and the strict policy positivity condition above holds. Define*

$$\Lambda_{N,\text{I-HJ}}(\delta) = \log\{20N_\infty(1/N, \widetilde{\Phi}_H^\star, 2N)/\delta\}.$$

Set $a_{N,H}(\delta) = \sqrt{18\Lambda_{N,\text{I-HJ}}(\delta)/N}$ and $b_{N,H}(\delta) = 16B_{\phi,H}\Lambda_{N,\text{I-HJ}}(\delta)/N$. With probability at least $1-\delta$, simultaneously for all $p \in \mathcal{P}$,

$$|\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)| \le \widehat{\alpha}_{N,\text{I-HJ}}(p), \tag{C.6}$$

$$\widehat{\alpha}_{N,\text{I-HJ}}(p) = a_{N,H}(\delta)\widetilde{\sigma}_{H,N}(p) + b_{N,H}(\delta). \tag{C.7}$$

Moreover,

$$N_\infty(1/N, \widetilde{\Phi}_H^\star, 2N) \le (N+1)N_\infty(1/(2Nn_{\max}), \mathcal{P}, 2Nn_{\max}). \tag{C.8}$$

Proof. Apply Theorem 6 of Maurer and Pontil (2009) to $\widetilde{\Phi}_H^\star$ and $1 - \widetilde{\Phi}_H^\star$, each at level $\delta/2$. The shift in $\widetilde{g}_{p,m}$ is constant across observations, so it affects neither deviations nor standard

deviations. Rescaling by $B_{\phi,H}$ and using $15/(N-1) \le 16/N$ gives, uniformly in (p, m),

$$|(P - P_N)g_{p,m}| \le a_{N,H}(\delta)s_N(g_{p,m}) + b_{N,H}(\delta). \tag{C.9}$$

Evaluate this uniform inequality at $m = \widehat{\bar{Y}}_{\text{I-HJ}}(p) \in [-M, M]$. Identity (C.3) and (C.5) give the claimed bound.

For the covering bound, an r-cover of $\mathcal{P}$ on the covariates of $2N$ clusters gives $|w_i(p) - w_i(q)| \le n_{\max} r/\eta$. Since $0 \le \eta w_i(q) \le 1$,

$$|\widetilde{g}_{p,m}(O_i) - \widetilde{g}_{q,m'}(O_i)| \le n_{\max} r + |m - m'|/(2M).$$

Take $r = 1/(2Nn_{\max})$ and an M/N-cover of $[-M, M]$ with at most $N + 1$ centers. Their product is the required $1/N$-cover. □

Lemma C.6 (Control of the Hájek variance-adaptive margin)**.** *Under the assumptions of Theorem C.2, with probability at least* $1 - \delta$*,*

$$\sup_{p\in\mathcal{P}} |s_N(\phi^{\text{I-HJ}}(p)) - \sigma_{\text{I-HJ}}(p)| \le 8B_{\phi,H}\sqrt{\Lambda_{N,\text{I-HJ}}(\delta)/N}. \tag{C.10}$$

If, additionally,

$$a_{N,H}(\delta) \sup_{p\in\mathcal{P}} \widehat{\sigma}_w(p) \le 1/2, \tag{C.11}$$

then, on the intersection of this event and the event of Theorem C.2, uniformly over p*,*

$$\widehat{\alpha}_{N,\text{I-HJ}}(p) \le 2\sqrt{18}\,\sigma_{\text{I-HJ}}(p)\sqrt{\Lambda_{N,\text{I-HJ}}(\delta)/N} + (16\sqrt{18} + 32)B_{\phi,H}\Lambda_{N,\text{I-HJ}}(\delta)/N.$$

Proof. Apply the normalized-class argument in the proof of Lemma C.5 to $\widetilde{\Phi}^{\star}_H$. Standard deviations are invariant under the m-dependent constant shift and scale by $B_{\phi,H}$. Since $\bar{Y}(p) \in [-M, M]$, the population-centered score belongs to $\mathcal{G}_H$, giving (C.10).

Write $a = a_{N,H}(\delta)$, $b = b_{N,H}(\delta)$ and $\delta_p = \widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)$. The sample standard deviation is a seminorm, so $\widetilde{\sigma}_{H,N}(p) \le s_N(\phi^{\text{I-HJ}}(p)) + |\delta_p|\widehat{\sigma}_w(p)$. On the Bernstein event, $|\delta_p| \le \widehat{\alpha}_{N,\text{I-HJ}}(p)$, and hence $\{1 - a\widehat{\sigma}_w(p)\}\widehat{\alpha}_{N,\text{I-HJ}}(p) \le a\, s_N(\phi^{\text{I-HJ}}(p)) + b$. Condition (C.11) gives $\widehat{\alpha}_{N,\text{I-HJ}}(p) \le 2a\, s_N(\phi^{\text{I-HJ}}(p)) + 2b$. Substitute (C.10) to conclude. □

C.2 Proof of Theorem 1

Proof. Recall the notation $R^{\star}_{G_0} = \inf_{p\in\mathcal{P}} R_{G_0}(p)$ and $V^{\star}_{\epsilon,G_0} = \inf_{p\in\mathcal{P}} V_{\epsilon,G_0}(p)$. Consider the decomposition

$$\begin{aligned} R_{G_0}(\hat{p}_{\text{I-IPW}}) - R^{\star}_{G_0} = \big(R_{G_0}(\hat{p}_{\text{I-IPW}}) - V_{\epsilon,G_0}(\hat{p}_{\text{I-IPW}})\big) + \big(V_{\epsilon,G_0}(\hat{p}_{\text{I-IPW}}) - V^{\star}_{\epsilon,G_0}\big) \\ + \big(V^{\star}_{\epsilon,G_0} - R^{\star}_{G_0}\big). \end{aligned} \tag{C.12}$$

Since $V_{\epsilon,G_0}(p) \geq R_{G_0}(p)$ for every $p \in \mathcal{P}$, the first term on the right-hand side is non-positive. Therefore,

$$R_{G_0}(\hat{p}_{\text{I-IPW}}) - R^\star_{G_0} \leq \underbrace{V^\star_{\epsilon,G_0} - R^\star_{G_0}}_{\text{approximation error}} + \underbrace{V_{\epsilon,G_0}(\hat{p}_{\text{I-IPW}}) - V^\star_{\epsilon,G_0}}_{\text{surrogate estimation error}}. \tag{C.13}$$

By empirical optimality of $\widehat{p}_{\text{I-IPW}}$ and the $2/\epsilon$-Lipschitz property of h_ϵ,

$$V_{\epsilon,G_0}(\widehat{p}_{\text{I-IPW}}) - V^\star_{\epsilon,G_0} \leq 2 \sup_{p\in\mathcal{P}} |V_{\epsilon,G_0}(p) - \widehat{V}_{\epsilon,\text{I-IPW}}(p)| \leq 2 \sup_{p\in\mathcal{P}} |\bar{C}(p) - \widehat{\bar{C}}(p)| + \frac{4C_0}{\epsilon} \sup_{p\in\mathcal{P}} |\bar{Y}(p) - \widehat{\bar{Y}}_{\text{I-IPW}}(p)|.$$

Taking expectations and applying Lemmas C.1 and C.2 gives

$$\mathbb{E}_{\mathcal{O}}[V_{\epsilon,G_0}(\widehat{p}_{\text{I-IPW}}) - V^\star_{\epsilon,G_0}] \leq 8 n_{\max} \mathcal{R}_N(\mathcal{P}) + \frac{8\sqrt{2} C_0 M}{\eta\epsilon} n_{\max}^{3/2} \mathcal{R}_N(\mathcal{P}) + \frac{4C_0 M}{\eta\epsilon} \sqrt{\frac{2\log 2}{N}}.$$

Combining this with (C.13) proves the theorem. □

C.3 Proof of Theorem 2

Proof. Recall the following

$$R_{G_0}(p) = \bar{C}(p) + C_0 \, \mathbf{1}\{\bar{Y}(p) < T\}, \qquad V_{\epsilon,G_0}(p) = \bar{C}(p) + C_0 \, h_\epsilon(\bar{Y}(p); T),$$

Note that $p^\star_\epsilon$ minimizes V_{ϵ,G_0} over $\mathcal{P}$. Since $p^\star$ solves the original constrained problem, $\bar{Y}(p^\star) \geq T$. For sufficiently small $\epsilon > 0$, we have $\epsilon/\kappa \leq t_0$. Set $\tilde{p} = p^\star + \frac{\epsilon}{\kappa} d$. By the growth condition,

$$\bar{Y}(\tilde{p}) \geq \bar{Y}(p^\star) + \kappa \frac{\epsilon}{\kappa} \geq T + \epsilon.$$

Hence $h_\epsilon(\bar{Y}(\tilde{p}); T) = 0$, so $V_{\epsilon,G_0}(\tilde{p}) = \bar{C}(\tilde{p})$., and by optimality of $p^\star_\epsilon$, $V^\star_{\epsilon,G_0} = V_{\epsilon,G_0}(p^\star_\epsilon) \leq V_{\epsilon,G_0}(\tilde{p}) = \bar{C}(\tilde{p})$. On the other hand, we have $R^\star_{G_0} = R_{G_0}(p^\star) = \bar{C}(p^\star)$,. Therefore,

$$V^\star_{\epsilon,G_0} - R^\star_{G_0} \leq \bar{C}_{G_0}(\tilde{p}) - \bar{C}_{G_0}(p^\star).$$

By Lipschitz continuity of $\bar{C}_{G_0}$,

$$\bar{C}_{G_0}(\tilde{p}) - \bar{C}_{G_0}(p^\star) \leq L_p \|\tilde{p} - p^\star\| = \frac{L_p \|d\|}{\kappa} \epsilon \implies V^\star_{\epsilon,G_0} - R^\star_{G_0} \leq \frac{L_p \|d\|}{\kappa} \epsilon.$$

□

C.4 Proof of Theorem 3

Proof. Define the following set

$$\mathcal{P}^+_N = \left\{ p \in \mathcal{P} : \bar{Y}(p) \geq T + \epsilon_N + B_{\text{I-IPW}}(\delta_N) \right\}.$$

Since $\epsilon_N + B_{\text{I-IPW}}(\delta_N) < s$, this set is nonempty. Further define the event

$$E_Y = \left\{ \sup_{p \in \mathcal{P}} \left| \widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p) \right| \leq B_{\text{I-IPW}}(\delta_N) \right\}.$$

By Lemma C.2, $\Pr(E_Y^c) \leq \delta_N$, where E_Y^c defines the complement event of E_Y. On E_Y, every $p \in \mathcal{P}_N^+$ satisfies

$$\widehat{\bar{Y}}_{\text{I-IPW}}(p) \geq \bar{Y}(p) - B_{\text{I-IPW}}(\delta_N) \geq T + \epsilon_N.$$

Therefore, applying Lemma C.4 with constant margin $m := \epsilon_N$ yields

$$\widehat{\bar{C}}(\hat{p}_{\text{I-IPW}}) \leq \inf_{p \in \mathcal{P}_N^+} \widehat{\bar{C}}(p), \qquad \widehat{\bar{Y}}_{\text{I-IPW}}(\hat{p}_{\text{I-IPW}}) \geq T + c\epsilon_N.$$

Since $c\epsilon_N \geq B_{\text{I-IPW}}(\delta_N)$, on E_Y we also have

$$\bar{Y}(\hat{p}_{\text{I-IPW}}) \geq \widehat{\bar{Y}}_{\text{I-IPW}}(\hat{p}_{\text{I-IPW}}) - B_{\text{I-IPW}}(\delta_N) \geq T + c\epsilon_N - B_{\text{I-IPW}}(\delta_N) \geq T.$$

Hence, the estimated policy satisfies the population outcome constraint on E_Y.

Moreover, on E_Y,

$$\bar{C}(\hat{p}_{\text{I-IPW}}) \leq \inf_{p \in \mathcal{P}_N^+} \bar{C}(p) + 2 \sup_{p \in \mathcal{P}} |\widehat{\bar{C}}(p) - \bar{C}(p)|.$$

Since $\bar{Y}(\hat{p}_{\text{I-IPW}}) \geq T$ on E_Y, it follows that

$$R_{G_0}(\hat{p}_{\text{I-IPW}}) - R^\star_{G_0} \leq \bar{C}^\star(T + \epsilon_N + B_{\text{I-IPW}}(\delta_N)) - \bar{C}^\star(T) + 2 \sup_{p \in \mathcal{P}} |\widehat{\bar{C}}(p) - \bar{C}(p)|.$$

On E_Y^c, regret is bounded by $1 + C_0$. Taking expectations and applying Lemma C.1 therefore gives

$$\text{Reg}(\hat{p}_{\text{I-IPW}}) \leq \bar{C}^\star(T + \epsilon_N + B_{\text{I-IPW}}(\delta_N)) - \bar{C}^\star(T) + 8 n_{\max} \mathcal{R}_N(\mathcal{P}) + (1 + C_0)\delta_N.$$

□

C.5 Proof of Theorem 4

Proof. For the first statement, the regret decomposition used in Theorem 1 and empirical optimality of $\widehat{p}_{\text{I-HJ}}$ give

$$\begin{aligned} R_{G_0}(\widehat{p}_{\text{I-HJ}}) - R^\star_{G_0} &\leq V^\star_{\epsilon, G_0} - R^\star_{G_0} + 2 \sup_{p \in \mathcal{P}} |V_{\epsilon, G_0}(p) - \widehat{V}_{\epsilon, \text{I-HJ}}(p)| \\ &\leq V^\star_{\epsilon, G_0} - R^\star_{G_0} + 2 \sup_{p \in \mathcal{P}} |\widehat{\bar{C}}(p) - \bar{C}(p)| + \frac{4C_0}{\epsilon} \sup_{p \in \mathcal{P}} |\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)|, \end{aligned}$$

where the second inequality uses the $2/\epsilon$-Lipschitz property of h_ϵ. Taking expectations and applying Lemmas C.1 and C.3 gives

$$\text{Reg}(\widehat{p}_{\text{I-HJ}}) \leq V^\star_{\epsilon, G_0} - R^\star_{G_0} + 8 n_{\max} \mathcal{R}_N(\mathcal{P}) + 24\sqrt{2} \frac{C_0 M}{\eta \epsilon} n_{\max}^{3/2} \mathcal{R}_N(\mathcal{P}) + \frac{4C_0 M}{\eta \epsilon \sqrt{N}} (2 + \sqrt{2 \log 2}),$$

proving the first statement.

For the second statement, let $B_H = B_{\text{I-HJ}}(\delta_N/2)$ and define $\mathcal{P}_N^+ = \{p \in \mathcal{P} : \bar{Y}(p) \geq T + \epsilon_N + B_H\}$ and $E_H = \{\sup_{p\in\mathcal{P}} |\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)| \leq B_H\}$. By Lemma C.3, $\Pr(E_H^c) \leq \delta_N/2$. Moreover, $\mathcal{P}_N^+$ is nonempty since $\epsilon_N + B_H < s$.

On E_H, every $p \in \mathcal{P}_N^+$ satisfies $\widehat{\bar{Y}}_{\text{I-HJ}}(p) \geq T + \epsilon_N$. Thus, Lemma C.4 gives $\widehat{\bar{C}}(\widehat{p}_{\text{I-HJ}}) \leq \inf_{p\in\mathcal{P}_N^+} \widehat{\bar{C}}(p)$ and $\widehat{\bar{Y}}_{\text{I-HJ}}(\widehat{p}_{\text{I-HJ}}) \geq T + c\epsilon_N$. Since $c\epsilon_N \geq B_H$, we also have $\bar{Y}(\widehat{p}_{\text{I-HJ}}) \geq T$ on E_H.

Furthermore, on E_H,

$$\bar{C}(\widehat{p}_{\text{I-HJ}}) \leq \inf_{p\in\mathcal{P}_N^+} \bar{C}(p) + 2\sup_{p\in\mathcal{P}} |\widehat{\bar{C}}(p) - \bar{C}(p)| = \bar{C}^\star(T + \epsilon_N + B_H) + 2\sup_{p\in\mathcal{P}} |\widehat{\bar{C}}(p) - \bar{C}(p)|.$$

Since $R^\star_{G_0} = \bar{C}^\star(T)$ and regret is at most $1 + C_0$ on E_H^c, taking expectations and applying Lemma C.1 yields

$$\text{Reg}(\widehat{p}_{\text{I-HJ}}) \leq \bar{C}^\star(T + \epsilon_N + B_H) - \bar{C}^\star(T) + 8n_{\max}\mathcal{R}_N(\mathcal{P}) + (1 + C_0)\delta_N.$$

Finally, if $\mathcal{R}_N(\mathcal{P}) = O(N^{-1/2})$, $\delta_N = N^{-1}$ and $\epsilon_N = B_H/c$, then $B_H = O(\sqrt{\log N/N})$ and $\epsilon_N + B_H \to 0$. Under the assumed local behavior of $\bar{C}^\star$, the preceding bound is $O(\sqrt{\log N/N})$, proving the second statement. □

C.6 Proof of Theorem 5

Proof. Write $L = \Lambda_N = \Lambda_{N,\text{I-IPW}}(\delta_N/2)$ and $B_\psi = M/\eta$. Since $\sigma_{\text{I-IPW}}(p) \leq B_\psi$ uniformly and $L/N \to 0$, $\sup_p \Delta_N(p) \to 0$. The positive slack assumption supplies a fixed policy with $\bar{Y}(p) > T$, so $\mathcal{A}_N^\Delta$ is nonempty for all sufficiently large N. Increase N_0 if necessary so that $N_0 \geq 16$.

Let $\hat{p} = \widehat{p}^{\text{VA}}_{\text{I-IPW}}$, and, for any $\xi > 0$, choose $q \in \mathcal{A}_N^\Delta$ with $\bar{C}(q) \leq \inf_{p\in\mathcal{A}_N^\Delta} \bar{C}(p) + \xi$. Define

$$\mathcal{E}_Y = \{|\widehat{\bar{Y}}_{\text{I-IPW}}(p) - \bar{Y}(p)| \leq c\widehat{\epsilon}_N(p)\ \forall p\}, \qquad \mathcal{E}_\epsilon = \{(1 + c)\widehat{\epsilon}_N(p) \leq \Delta_N(p)\ \forall p\}.$$

Theorem C.1 gives $\Pr(\mathcal{E}_Y^c) \leq \delta_N/2$. By Lemma C.5, with probability at least $1 - \delta_N/2$, $c\widehat{\epsilon}_N(p) \leq \sqrt{18}\,\sigma_{\text{I-IPW}}(p)\sqrt{L/N} + (16\sqrt{18}+32)B_\psi L/N$ uniformly in p. Since $d_1 \geq (1+c^{-1})\sqrt{18}$ and $d_2 \geq (1 + c^{-1})(16\sqrt{18} + 32)$, the event $\mathcal{E} = \mathcal{E}_Y \cap \mathcal{E}_\epsilon$ has probability at least $1 - \delta_N$.

On $\mathcal{E}$, $\widehat{\bar{Y}}_{\text{I-IPW}}(q) \geq \bar{Y}(q) - c\widehat{\epsilon}_N(q) \geq T + \widehat{\epsilon}_N(q)$. Lemma C.4, applied with $m(p) = \widehat{\epsilon}_N(p)$, gives $\widehat{\bar{C}}(\hat{p}) \leq \widehat{\bar{C}}(q)$ and $\widehat{\bar{Y}}_{\text{I-IPW}}(\hat{p}) \geq T + c\widehat{\epsilon}_N(\hat{p})$. Thus $\bar{Y}(\hat{p}) \geq \widehat{\bar{Y}}_{\text{I-IPW}}(\hat{p}) - c\widehat{\epsilon}_N(\hat{p}) \geq T$ on $\mathcal{E}$, and $\Pr\{\bar{Y}(\hat{p}) < T\} \leq \delta_N$.

Let $Z_C^\pm = \sup_p\{\pm(\widehat{\bar{C}}(p) - \bar{C}(p))\}$. On $\mathcal{E}$, $\bar{C}(\hat{p}) - \bar{C}(q) \leq Z_C^+ + Z_C^-$. On $\mathcal{E}^c$, the cost difference is at most one. The sum $Z_C^+ + Z_C^-$ is always nonnegative; the two one-sided bounds in the proof of Lemma C.1 give $\mathbb{E}_\mathcal{O}(Z_C^+ + Z_C^-) \leq 4n_{\max}\mathcal{R}_N(\mathcal{P})$. Consequently,

$$\mathbb{E}_\mathcal{O}\{\bar{C}(\hat{p}) - \bar{C}(q)\} \leq 4n_{\max}\mathcal{R}_N(\mathcal{P}) + \delta_N.$$

As in Theorem 3, nonempty population feasibility gives $R^\star_{G_0} = \bar{C}^\star(T)$. Add the constraint penalty $C_0 \Pr\{\bar{Y}(\hat{p}) < T\}$ to the preceding bound and let $\xi \downarrow 0$. This proves the displayed regret inequality. The final rate follows from the additional conditions in the theorem. □

C.7 Proof of Theorem C.3

Use the preliminary results above and evaluate $\widehat{\alpha}_{N,\text{I-HJ}}$ at $\delta = \delta_N/2$. For d_1, d_2 depending only on C_0, define

$$\Delta_{N,\text{I-HJ}}(p) = d_1 \sigma_{\text{I-HJ}}(p) \sqrt{\Lambda_{N,\text{I-HJ}}(\delta_N/2)/N} + d_2 B_{\phi,H} \Lambda_{N,\text{I-HJ}}(\delta_N/2)/N. \tag{C.14}$$

Let $\mathcal{A}^{\Delta}_{N,\text{I-HJ}} = \{p \in \mathcal{P} : \bar{Y}(p) \geq T + \Delta_{N,\text{I-HJ}}(p)\}$ and, when this set is nonempty, $\bar{C}^{\star,\text{VA}}_{N,\text{I-HJ}} = \inf_{p \in \mathcal{A}^{\Delta}_{N,\text{I-HJ}}} \bar{C}(p)$.

Since $0 \leq w_i(p) \leq 1/\eta$, $\sup_p \widehat{\sigma}_w(p) \leq \eta^{-1}\sqrt{N/(N-1)}$. Hence (C.11), evaluated at $\delta_N/2$, is automatic for all sufficiently large N if $\Lambda_{N,\text{I-HJ}}(\delta_N/2) = o(N)$. Take $d_1 \geq (1 + c^{-1})2\sqrt{18}$ and $d_2 \geq (1 + c^{-1})(16\sqrt{18} + 32)$. Theorem C.2 and Lemma C.6, each at level $\delta_N/2$, then imply, with probability at least $1 - \delta_N$,

$$|\widehat{\bar{Y}}_{\text{I-HJ}}(p) - \bar{Y}(p)| \leq \widehat{\alpha}_{N,\text{I-HJ}}(p), \qquad (1 + c^{-1})\widehat{\alpha}_{N,\text{I-HJ}}(p) \leq \Delta_{N,\text{I-HJ}}(p) \quad \text{for all } p \in \mathcal{P}. \tag{C.15}$$

Here positivity follows directly from the strict policy positivity condition of Lemma C.3.

Theorem C.3 (Variance-adaptive Hájek refinement)**.** *Suppose Assumptions 1–3 hold, $N \geq 16$, $\delta_N \to 0$, and $0 < p(x,v) < 1$ for all $p \in \mathcal{P}$ and covariate pairs (x,v). Suppose also that (C.11) holds at level $\delta_N/2$ for every sample and $\mathcal{A}^{\Delta}_{N,\text{I-HJ}} \neq \varnothing$. Set $\widehat{\epsilon}_{N,\text{I-HJ}}(p) = \widehat{\alpha}_{N,\text{I-HJ}}(p)/c$, evaluated at $\delta = \delta_N/2$, and let*

$$\widehat{p}^{\text{VA}}_{\text{I-HJ}} \in \arg\min_{p \in \mathcal{P}} \{\widehat{\bar{C}}(p) + C_0 h_{\widehat{\epsilon}_{N,\text{I-HJ}}(p)}(\widehat{\bar{Y}}_{\text{I-HJ}}(p); T)\}.$$

Then

$$\mathbb{E}_{\mathcal{O}}\{R_{G_0}(\widehat{p}^{\text{VA}}_{\text{I-HJ}}) - R^\star_{G_0}\} \leq \bar{C}^{\star,\text{VA}}_{N,\text{I-HJ}} - \bar{C}^\star(T) + 4n_{\max}\mathcal{R}_N(\mathcal{P}) + (1 + C_0)\delta_N. \tag{C.16}$$

Proof. By (C.15), the two uniform events used in the proof of Theorem 5 hold for the Hájek estimator with probability at least $1 - \delta_N$, using $\widehat{\epsilon}_{N,\text{I-HJ}}(p)$ in place of $\widehat{\epsilon}_N(p)$ and $\Delta_{N,\text{I-HJ}}$ in place of Δ_N. Apply the same argument with $\widehat{\bar{Y}}_{\text{I-HJ}}$ in place of $\widehat{\bar{Y}}_{\text{I-IPW}}$. In particular, every $q \in \mathcal{A}^{\Delta}_{N,\text{I-HJ}}$ has zero empirical penalty on this event; Lemma C.4 gives $\widehat{\bar{C}}(\hat{p}) \leq \widehat{\bar{C}}(q)$ and $\widehat{\bar{Y}}_{\text{I-HJ}}(\hat{p}) \geq T + \widehat{\alpha}_{N,\text{I-HJ}}(\hat{p})$, hence population feasibility. The cost argument is unchanged and gives $4n_{\max}\mathcal{R}_N(\mathcal{P}) + \delta_N$; the constraint penalty contributes $C_0\delta_N$. Taking the infimum over comparison policies proves the result. □

C.8 Proof of Theorem A.1

Proof. The functional $\bar{C}$ is 1-Lipschitz in the sup norm: $|\bar{C}(p')-\bar{C}(p)| \le \|p'-p\|_\infty$. Suppose a budget-constrained maximizer $p^\star$ satisfies $\bar{C}(p^\star) < B$. By local outcome growth, a policy p' arbitrarily close to $p^\star$ has strictly higher welfare. Taking $\|p'-p^\star\|_\infty < B - \bar{C}(p^\star)$ keeps p' within budget, a contradiction. Thus every budget-constrained maximizer has cost exactly B. Any such maximizer attains $T^\star$ at cost B, so $\bar{C}^\star(T^\star) \le B$. If $\bar{C}^\star(T^\star) < B$, there is a policy q with $\bar{C}(q) < B$ and $\bar{Y}(q) \ge T^\star$. Such a policy is also a budget-constrained maximizer but does not exhaust the budget, contradicting the preceding argument. Hence $\bar{C}^\star(T^\star) = B$. A policy optimal for either formulation is feasible and optimal for the other, proving equality of the optimizer sets. □

C.9 Proof of Theorem A.2

Proof. For part (i), write $V_\epsilon^{W,\star} = \sup_p V_\epsilon^W(p)$. Because $h_\epsilon^+(c;B) \ge \mathbf{1}\{c > B\}$, $V_\epsilon^W(p) \le R^W(p)$ for all p. Further, every policy with $\bar{C}(p) \le B-\epsilon$ has zero hinge penalty, so $V_\epsilon^{W,\star} \ge \bar{Y}^\star(B-\epsilon)$. Empirical optimality and the $2/\epsilon$-Lipschitz property of the hinge imply

$$\begin{aligned}\bar{Y}^\star(B) - R^W(\widehat{p}_\epsilon^W) &\le \bar{Y}^\star(B) - V_\epsilon^{W,\star} + 2\sup_{p\in\mathcal{P}} |\widehat{V}_\epsilon^W(p) - V_\epsilon^W(p)| \\ &\le \bar{Y}^\star(B) - \bar{Y}^\star(B-\epsilon) + 2\sup_p |\widehat{\bar{Y}}^{\text{I-IPW}}(p) - \bar{Y}(p)| + \frac{4C_y}{\epsilon}\sup_p |\widehat{\bar{C}}(p) - \bar{C}(p)|.\end{aligned}$$

Taking expectations and applying the expected uniform deviations in Lemmas 1 and 2 proves (A.7).

For part (ii), let E_C and E_Y be the uniform-deviation events $\sup_p |\widehat{\bar{C}}(p) - \bar{C}(p)| \le b_C(\delta)$ and $\sup_p |\widehat{\bar{Y}}^{\text{I-IPW}}(p) - \bar{Y}(p)| \le b_Y(\delta)$. By Lemmas C.1 and C.2 and a union bound, $\Pr(E_C \cap E_Y) \ge 1-\delta$. Fix any $q \in \mathcal{P}$ with $\bar{C}(q) \le B - \epsilon - b_C(\delta)$. On E_C, $\widehat{\bar{C}}(q) \le B-\epsilon$, so its empirical hinge penalty is zero. Empirical optimality therefore gives

$$C_y h_\epsilon^+\{\widehat{\bar{C}}(\widehat{p}_\epsilon^W); B\} \le \widehat{\bar{Y}}^{\text{I-IPW}}(\widehat{p}_\epsilon^W) - \widehat{\bar{Y}}^{\text{I-IPW}}(q).$$

On E_Y, the right-hand side is bounded above by $2M + 2b_Y(\delta) = C_y\alpha_N(\delta)$. By the definition of the reflected hinge, the inequality $h_\epsilon^+(c;B) \le \alpha < 1$ implies $c \le B - \epsilon k(\alpha)$. Consequently,

$$\widehat{\bar{C}}(\widehat{p}_\epsilon^W) \le B - \epsilon k\{\alpha_N(\delta)\} \le B - b_C(\delta),$$

so $\bar{C}(\widehat{p}_\epsilon^W) \le B$ on $E_C \cap E_Y$. The same empirical optimality inequality, and the nonnegativity of the penalty, imply $\widehat{\bar{Y}}^{\text{I-IPW}}(\widehat{p}_\epsilon^W) \ge \widehat{\bar{Y}}^{\text{I-IPW}}(q)$. Thus, on $E_C \cap E_Y$,

$$\bar{Y}(\widehat{p}_\epsilon^W) \ge \bar{Y}(q) - 2b_Y(\delta).$$

Taking the supremum over all such q and using the population budget feasibility just shown

gives

$$\bar{Y}^\star(B) - R^W(\widehat{p}_\epsilon^W) \leq \bar{Y}^\star(B) - \bar{Y}^\star\{B - \epsilon - b_C(\delta)\} + 2b_Y(\delta).$$

On the complement of $E_C \cap E_Y$, the regret is bounded above by $2M + C_y$. Taking expectations yields (A.8). The rate follows because $b_C(\delta_N)$ and $b_Y(\delta_N)$ are $O\{\sqrt{\log N/N}\}$, while $k\{\alpha_N(\delta_N)\}$ converges to $k(2M/C_y) > 0$. □

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